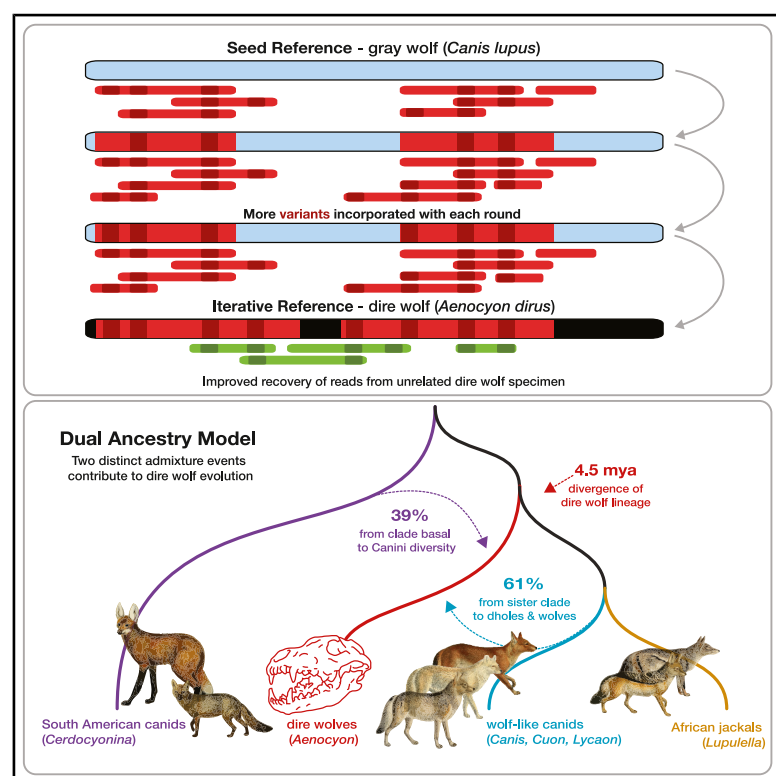


On the ancestry and evolution of the extinct dire wolf

Graphical abstract



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In brief

Gedman, Morrill-Pirovich, et al. reconstruct the most complete and high-coverage dire wolf genome to date from ancient DNA using iterative and pangenome approaches. The authors find evidence supporting a new model of dire wolf evolution with two distinct canid lineages contributing ancestry to this iconic Pleistocene carnivore.

Highlights

- Paleogenomes (12.8×, 3.4×) generated from dire wolf specimens (>72,000, ~13,000 ya)
- Nuclear genome reconstruction via iterative mapping reduces reference bias
- Dire wolf lineage split from jackals and wolf-like canids 4.5 mya
- Dual ancestry model supports admixture in evolution of dire wolves

Short article

On the ancestry and evolution of the extinct dire wolf

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SUMMARY

Dire wolves (*Aenocyon dirus*) are extinct predators of Pleistocene North America. Although phenotypically similar to living wolves (*Canis lupus*), dire wolves have yet to be placed confidently in the canid family tree. We generated 3.4× and 12.8× paleogenomes from two well-preserved dire wolves dating to >13,000 and >72,000 years ago and estimated consensus species trees for these and 10 canid species. Our results revealed that ~2/3 of dire wolf ancestry is most likely derived from a lineage sister to the clade comprising the gray wolf, coyote, and dhole and the remaining ~1/3 from a lineage near the base of Canini diversity. Our results underscore the power of paleogenomes to resolve long-standing taxonomic questions and contribute to growing evidence of the role of post-speciation gene flow as an evolutionary force.

INTRODUCTION

The dire wolf (*Aenocyon dirus*) first appeared in the North American fossil record during the late Middle Pleistocene, ~250,000 years before present (ybp),¹ and was the most prominent canid

species in continental North America until its extinction ~12,900 ybp (radiocarbon calibrated).² Today, dire wolf fossils are ascribed to two subspecies based on size and geography: the smaller ~60 kg *A. d. guildayi* in the west and the larger ~68 kg *A. d. dirus* in the east.^{3,4} Morphologically, dire wolves

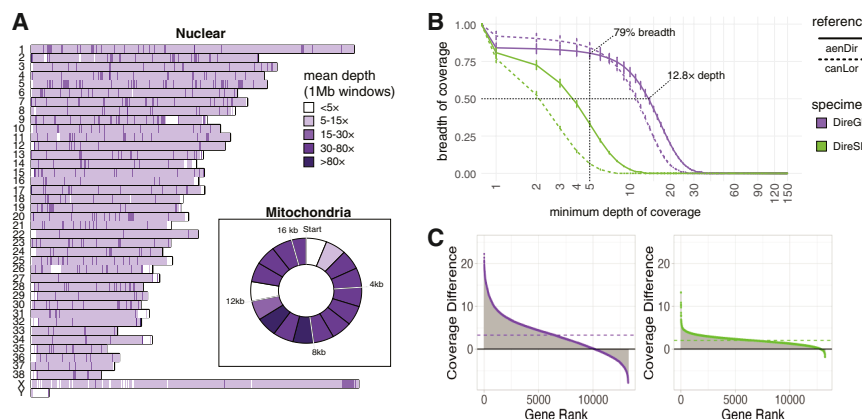


Figure 1. Iterative paleogenome inference improves whole-genome and -exome mappability

(A) Mean read depth of coverage by DireGB libraries mapped over 1 Mb windows from the reference assembly of *A. dirus* reconstructed by iterative mapping and polishing. The inset displays mean read depth over the reconstructed full-length mitochondrial sequence, split by 1 Kb bins.

(B) Cumulative coverage of DireGB (purple) and DireSP (green) libraries mapped to divergent gray wolf reference “canLor” (dotted line) and iterative dire wolf reference “aenDir” (solid line) plotting mean and standard deviation in proportion of bases covered across 38 autosomes. The proportion of the reference genome (i.e., breadth of coverage) at 5× and average depth for DireGB against the dire wolf reference is highlighted.

(C) Average coverage difference over annotated orthologous genes ($n = 13,230$) relative to dire wolf (positive) and gray wolf (negative). Most genes showed improved average coverage (dotted line) using aenDir in both DireGB (77% + 3.3×) and DireSP (97% + 2.1×).

were similar to gray wolves (*Canis lupus*), with larger temporalis muscles, suggesting a greater bite force,⁵ and proportionately longer bacula, suggesting sexual competition.^{6,7}

Dire wolves originated in North America,^{8–10} but their evolutionary history and taxonomic relationships to other canid species remain uncertain. A recent analysis of ~0.01×- to 0.23×-coverage dire wolf genomes concluded that the dire wolf lineage first diverged from wolf-like canids during the late Miocene, ~6 million years ago (mya).¹¹ This timing coincided with the appearance in the Eurasian and African fossil records of the genus *Eucyon*, the lineage from which extant wolf-like canid species probably descended.¹² During the Pleistocene, shifting habitats and the ephemeral emergence of the Bering Land Bridge provided repeated opportunities for wolf-like lineages to disperse in and out of continental North America. Despite the geographic and temporal overlap between the ranges of dire wolves, coyotes, and gray wolves, no evidence of admixture between dire wolves and either lineage has been identified.¹¹ Since admixture between closely related canid species is common,^{13–15} the absence of gene flow between dire wolves and sympatric canids suggests temporal, geographic, reproductive, or behavioral isolation.

Although Perri et al.¹¹ confirmed that dire wolves are a distinct evolutionary lineage, phylogenetic incongruence among estimated gene trees led to uncertainties about their early evolutionary history. Gene trees showed conflicting topologies, with ~50% supporting dire wolves as the earliest-diverging wolf-like canid lineage and the remainder placing jackals as an outgroup to dire wolves and other wolf-like canids.² Divergence time estimates suggested the dire wolf lineage emerged after divergence of the South American *Cerdocyonina* subtribe 5–4 mya^{11,16} and before divergence of the common ancestors of African jackals (*Lupulella* spp.) 7.6–3.5 My and wolf-like canids 8.5–4.0 mya (Canina subtribe). Incomplete lineage sorting¹⁷ resulting from this rapid diversification could be one driver of the observed tree incongruence, although gene flow from unsampled populations could not be excluded.

Resolving the phylogenetic position of dire wolves and assessing evidence for gene flow is hindered by the deep

divergence among canids and the technical challenges of analyzing ancient DNA. In addition to damage, ancient DNA is susceptible to reference bias, where divergent reads fail to map, artificially inflating similarity to the reference¹⁸ and influencing phylogenetic and demographic inference.^{19,20} This bias is particularly problematic when the reference genome is evolutionarily distant from the paleogenome,^{21,22} as in the case of the dire wolf.

To better understand the origin and evolution of dire wolves, we leveraged improved paleogenomics approaches to recover additional ancient DNA from two well-preserved dire wolves¹¹: DireSP, a 13,000-year-old incisor from Sheriden Pit, Ohio,²³ and DireGB, a $\geq 72,000 \pm 14,000$ -year-old skull from Gigantobison Bay, Idaho.²⁴ We generated paleogenomes for both specimens and performed phylogenetic and admixture analyses that provide new insights into the evolution of the dire wolf.

RESULTS

We generated paleogenomes with 3.4× depth of coverage for DireSP and 12.8× coverage for DireGB. After filtering, trimming, and merging overlapping reads, we retained 2.8 billion unique sequence reads for DireSP and 33.3 billion unique sequence reads for DireGB (Table S1). We mapped reads using BWA *aln* against the Greenland gray wolf chromosome-level (including Y) genome assembly (GenBank: GCA_905319855.2, also known as “canLor”)²⁵ (Table S2). After mapping, 6% of the genome was covered >5× for DireSP and 82% for DireGB (Figures 1A and 1B; Table S3). Previously, ancient DNA isolated from these same individuals yielded 0.2× (DireSP) and 0.3× (DireGB) coverage¹¹ (Table S3). We found patterns of DNA damage expected for single-stranded libraries (5′ C>T: DireGB = 30.2%; DireSP = 26.6%; Figure S1). Mapping to the gray wolf reference sex chromosomes versus autosomes yielded Rx ratios^{26,27} indicative of male individuals (95% confidence intervals [CIs]: DireSP = 0.539–0.554 and DireGB = 0.538–0.554).

We used an iterative process to reconstruct a progressively less divergent nuclear paleogenome reference (“aenDir”). After six iterations, we identified ~14.6 million changes (13,665,448

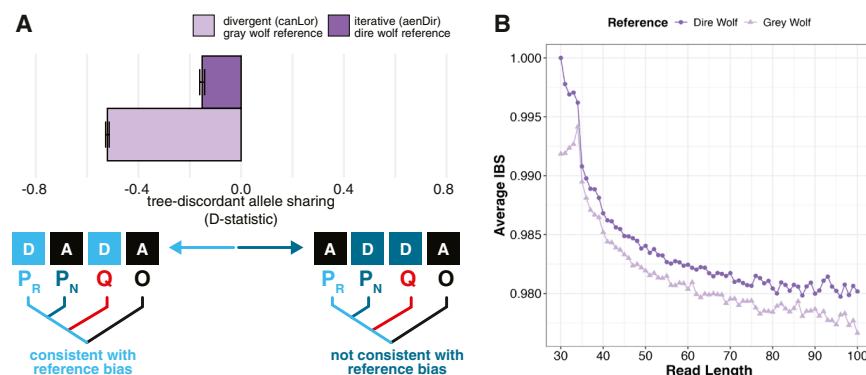


Figure 2. The iterative genome reconstruction process reduces reference bias

(A) Mapping to aenDir reduced reference bias 0.7- to 0.8-fold compared to mapping to canLor. Residual reference bias was assessed using the *D*-statistic to measure derived allele sharing between the DireGB consensus genome called against the two seed references. Ancestral (A) and derived (D) alleles were localized by aligning either of the query (*A. dirus*; red, Q) consensus DireGB sequences (canLor or aenDir called), seed reference Greenland gray wolf (*C. lupus orion* in light blue, P_R) and non-seed reference North American gray wolf (*C. lupus*; dark blue, P_N), and outgroup red fox (*V. vulpes*; black, O) to a common reference genomic coordinates from a gray fox

(*U. cinereoargenteus*). Error bars represent standard errors from the block jackknife procedure to divide the outgroup genomes into 1 Mb, equally spaced, contiguous blocks.

(B) Average identity by state (IBS) by read length for DireSP reads mapped to either the DireGB-based iteratively reconstructed paleogenome (dark purple) or divergent gray wolf (light purple) reference assemblies. IBS is defined as ((matches – mismatch)/aligned read length).

single-nucleotide variants, 388,514 insertions, and 515,129 deletions) from the seed reference assembly (gray wolf). We then called a pseudo-haploid, variant-based consensus sequence that represented the highest-depth alleles. In parallel, we used established methods^{21,28–30} to reconstruct the mitochondrial paleogenome at high coverage (average depth = 158.4×). Merging both nuclear and mitochondrial reconstructed paleogenomes resulted in a final reference assembly with a hard-masked N content of 16.84% (Table S4).

Mapping ancient DNA libraries to the reconstructed paleogenome reference (aenDir) increased coverage at both the whole-genome and -exome levels (Figures 1A and 1B). Compared with the initial mapping to the gray wolf reference (canLor), mapping to aenDir resulted in an increase in average coverage of 2.8× for DireGB and 1.5× for DireSP libraries. Annotation of aenDir identified the majority of protein-coding genes (71%) in canLor, with most missing genes due to masking of low-coverage regions (Figures S2 and S3). The depth of coverage in exons improved by ~2-fold for both DireSP and DireGB when mapped against aenDir (Figure 1C; Table S5).

Mapping against aenDir also reduced the impact of reference bias (Figure 2A; Table S6). We found a significant decrease in allele sharing for the DireGB consensus called against aenDir compared to canLor, estimated using the *D*-statistic (seed reference *C. lupus*, non-seed reference *C. lupus*; consensus *A. dirus*, and outgroup *Vulpes vulpes*) (*D* for aenDir = −0.152, *Z* score = −16.01; *D* for canLor = −0.521, *Z* score = −76.51). In a complementary test, we assessed the identity by state (IBS) for DireSP reads mapped against aenDir (Figure 2B). We found that DireSP reads mapped to aenDir exhibited higher IBS (i.e., greater sequence identity) than canLor across all read lengths, especially for reads <35 bp, suggesting the presence of residual reference bias, predominantly in shorter reads.

Dire wolf paleogenomes were more similar to extant Canina genomes than anticipated, given a 5.8 mya divergence time,¹¹ using a mutation rate calibrated for *Canis* species³¹ and a wide range of generation times³² for priors (Figure S4). We computed sequence divergence on fixed autosomal transversions at sites covered at 5× relative to the total non-N sequence length of aenDir

(2,310,286,013 bp). We found the dire wolf paleogenomes less divergent (on average, 0.09% divergent) than expected based on prior parameters (0.6%–0.9% sequence divergence) and estimated sequence similarities of 99.91% to jackals and 99.94% to non-seed reference gray wolves at either end of wolf-like canid diversity (Figure S4A). As residual reference bias in mapping could inflate sequence similarity, we also assessed the distribution of unique 32-mers from contamination-filtered sequence reads using *mask* to estimate the genome distance.³³ We found that dire wolves exhibited equally low 32-mer diversity compared to gray wolf and jackal species (~0.01), with greater distance to red and gray foxes (0.02–0.03) (Figure S4B; Table S7).

Our estimated mitochondrial and nuclear phylogenies were similar to those published previously (Figures 3 and S5–S9; Tables S8, S9, and S10). The mitochondrial phylogeny showed the known mitonuclear discordance among canid lineages.¹⁵ Our nuclear phylogeny, which was based on 19.5 Mb of non-coding dire wolf sequence at least 1 Mb away from protein-coding genes and <25% missingness on average (Figures S6 and S7), placed dire wolves into a monophyletic clade that diverges prior to the split between the black-backed jackal and wolf-like canids (Figures 3 and S8). However, this placement was supported by only 44% of input loci, while 39% of loci supported a closer relationship between dire wolves and non-Lupulellan wolf-like canids (Figure 3). Divergence time estimates (Figure S9; Table S10) suggested that the dire wolf diverged from other canids ~4.5 mya (95% CI: 5.84–2.89 mya) and the jackals diverged ~4.08 mya (95% CI: 5.34–2.58 mya).

The discordance among estimated gene tree topologies arises from admixed ancestry in dire wolves (Figure 4). Using alignments to a canid pangenome to minimize reference bias (Figure S10), we find that dire wolves shared an excess of derived alleles with gray wolves relative to other canids, including dholes (11.9 ≤ *Z* ≤ 15.3), African wild dogs (61.9 ≤ *Z* ≤ 65.4), African jackals (92.3 ≤ *Z* ≤ 99.6), and South American canids (173.4 ≤ *Z* ≤ 183.7) (Figure 4A). We did not detect similar affinity between dire wolves and gray wolves when comparing to wolf-like canids more closely related gray wolves, including coyotes and red wolves (|*Z*| ≤ 2.4) (Figure 4A). We also observed

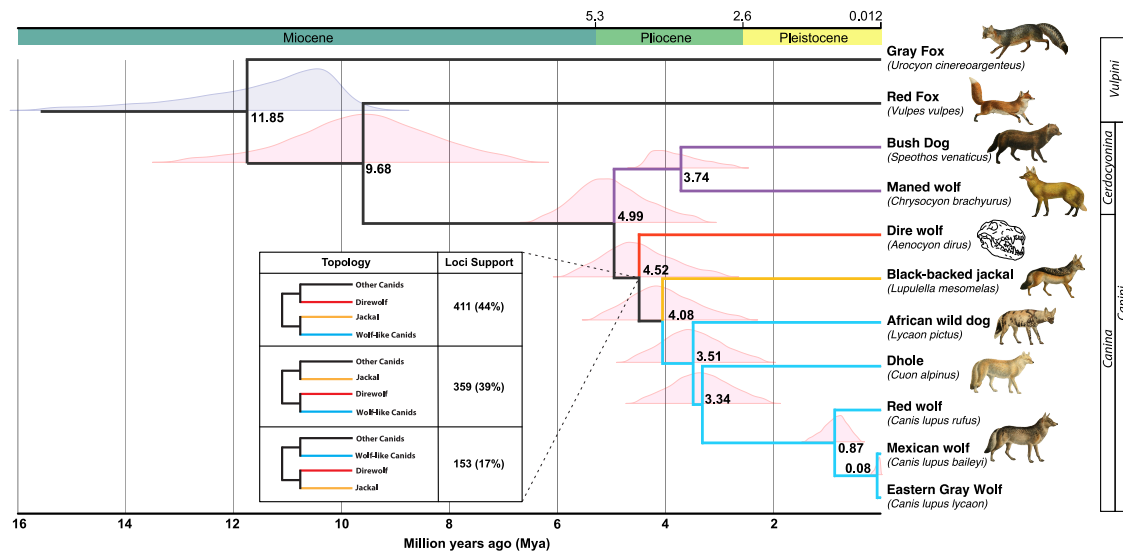


Figure 3. The dire wolf lineage is distinct from jackals and other wolf-like canids

Time-calibrated nuclear phylogeny from 924 loci (25 kb each) showing dire wolves diverging prior to the jackal/wolf-like canid split. Node distributions with fossil calibrations are shown in purple. The inset details the alternative topologies for the dire wolf node, along with the proportion of loci supporting each alternative.

that, relative to dire wolves, African jackals ($66.0 \leq Z \leq 72.4$) and other wolf-like canids ($Z \geq 92.7$) share more derived alleles with gray wolves (Figure 4B). Together, these patterns suggest that dire wolves had a mixture of ancestries from two distinct lineages, with one diverging prior to African jackals and another before gray wolves and coyotes. Allele sharing between dire wolves and South American canids relative to extant wolf-like canids ($-7.2 \leq Z \leq -2.3$) is consistent with the deeper ancestry component falling within Cerdocyonina diversity (Figure 4B). Both dire wolves formed a clade relative to all individual coyotes and gray wolves (Figure S10), implying limited gene flow between dire wolves and specific *Canis* lineages following their divergence.

Admixture graph models of canid population history identified both wolf-like canid and deep-ancestry components in the dire wolf lineage (Figure 4C). We applied a heuristic algorithm³⁴ to find optimal admixture graphs modeling the relationships between seven canids (dire wolves, Eurasian gray wolves, coyotes, dholes, African wild dogs, and maned wolves, using red foxes as an outgroup). No model permitting zero or one admixture event was supported by the data; however, the best-fitting model featuring two admixture events had statistical support (worst residual of $Z = 2.6$) (Figure 4C). This model found that dire wolves received ~39% of their ancestry from a lineage that groups with South American canids, with the remaining ~61% attributed to a sister lineage to dholes, coyotes, and gray wolves. This wolf-like ancestor may have been the source of the phylogenetically discordant dire wolf mitochondrial genome, which falls within wolf-like canid diversity. As suggested by evidence of excess derived allele sharing between dire wolves and South American canids, a similar model that places the minor ancestry component of dire wolves outside extant Canini diversity instead of on the South American canid branch provides a worse fit (worst residual of $Z = 5.25$), although the branch placing this line-

age within Cerdocyonina has a short drift edge. When analyzed separately, both dire wolf individuals produced similar optimal graphs (Figure S11), suggesting that admixture was a feature of dire wolf population history. Graphs including both an additional outgroup (gray foxes) and additional South American canid species confirmed that the divergent portion of dire wolf ancestry falls near the base of Canini diversity (Figure S12).

The phylogenetic placement of the wolf-like component in dire wolves is similar to a proposed unsampled canid lineage that contributed ancestry to the common ancestor of coyotes and wolves. The best-fitting admixture graph also features gene flow from dire wolves into wolves and coyotes, which could be the source of this unsampled ancestry. However, scenarios in which dholes, rather than coyotes and wolves, are admixed also provide similar fits to the data, and we observed that relationships among extant wolf-like canids can be modeled without requiring this unsampled admixture (Figure S13). Models of canid population history that include additional canid lineages highlight other known and cryptic admixture events and identify the gene flow event leading to dire wolves (Figure S14). We found mixed support for introgression from the dhole lineage into the African wild dog lineage, which could reflect cryptic admixture,³⁵ and consistent support for other previously detected admixture events.^{13,36,37} These findings reinforced that this approach is sensitive to detecting hybridization across diverse canids (Figure S15).

We tested models with varying complexity to assess support for dual ancestry in dire wolves. The simplest supported model required two admixture events (Figure 4C), but models with three admixture events also fit the data, yielding a broader range of graph topologies, including scenarios where lineages other than dire wolves are admixed (Table S11; Figure S14). We compared alignment- and genotype-calling strategies to assess their impact on admixture inference. Consensify-style³⁸ pseudohaploid genotype calls from pangenome alignments showed reduced biases

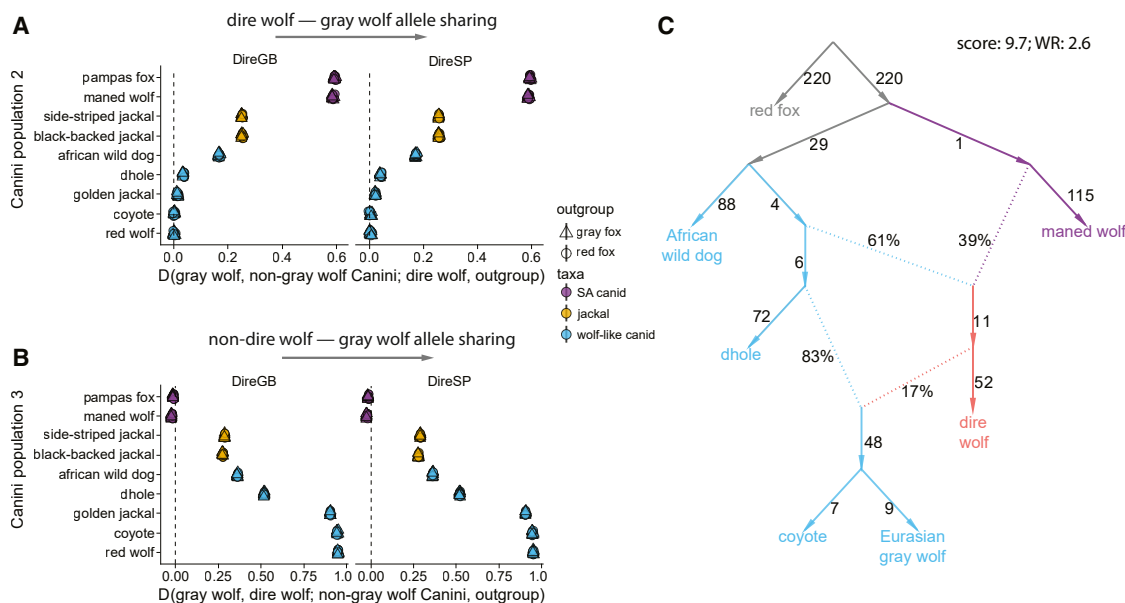


Figure 4. Investigating gene flow between dire wolves and other canid lineages

(A) D -statistics of the form $D(\text{gray wolf, other Canini group; dire wolf, red fox})$, investigating allele sharing between dire wolves and gray wolves, relative to other species and subspecies within Canini.

(B) D -statistics of the form $D(\text{gray wolf, dire wolf; other Canini group, red fox})$, investigating allele sharing between dire wolves and other populations within Canini, relative to gray wolves.

Bars depict 3 standard errors in (A) and (B). Each point is a comparison with a different individual gray wolf, with points colored by the Canini clade indicating the non-gray-wolf population in the comparison.

(C) Best-fitting admixture graph with two admixture events modeling the relationship between dire wolves ($n = 2$); wolf-like canids: Eurasian wolves ($n = 2$), coyotes ($n = 4$), dholes ($n = 2$), and African wild dogs ($n = 3$); and maned wolf ($n = 2$), with red fox ($n = 2$) as the outgroup. Additional graph fits using other taxa and alignment- and genotype-calling strategies are summarized in Table S11.

relative to other approaches (Figures S16 and S19), though this method requires greater read agreement, which could amplify mapping artifacts in low-coverage regions. We observed differences in allele frequency estimates and phylogenetic signals across reference alignment strategies (Figures S17–S19).

DISCUSSION

Dire wolf evolutionary history

The Late Pleistocene dire wolf descends from an admixed canid lineage formed by hybridization between two ancestral populations: one along the branch leading to the South American canids (Cercyonina) after their divergence from African jackals and other wolf-like canids (Canina) and a more recently derived wolf-like lineage that diverged after the emergence of African wild dogs (*Lycaon*) (Figures 3 and 4C). This dual ancestry explains the nuclear gene tree discordance observed here and previously¹² and is supported by analyses using both our iteratively polished dire wolf reference and a canid pangenome, demonstrating robustness across alignment strategies.

The primary difference between phylogenies estimated using these two mapping approaches was the placement of the dire wolves' deep minor ancestral component, with the pangenome approach placing this ancestry component within South American canid diversity and the iterative approach placing it outside of Canini. We attribute this discordance in part to reference bias

in the outgroup red fox alignments for the polished dire wolf reference, which pulls the deep ancestry component toward this group. These results suggest that the major dire wolf ancestor diverged after the *Lycaon* lineage, possibly from extinct *Canis* populations that expanded back into North America, supplanting or admixing with local populations.¹²

The ubiquity of gene flow among canids^{13,15,35–37} challenges inference of the precise placement of admixture events among these taxa. While our simplest supported model requires two admixture events to explain dire wolf ancestry (Figure 4C), more complex models with three admixture events also fit the data and yield alternative scenarios, including graphs where other lineages are admixed (Table S11). Given this extensive hybridization history, the true population history is likely more complex than any single model can capture. Nevertheless, our conclusion regarding the dual ancestry of dire wolves is robustly supported across different models, alignment strategies, and levels of complexity.

Excess basal ancestry in wolves and coyotes relative to other wolf-like canids was previously attributed to gene flow with an unknown wolf-like lineage that emerged after African wild dog divergence.¹⁵ Our results suggest this wolf-like lineage within dire wolves also contributed ancestry to wolves and coyotes, possibly through dire wolf-mediated gene flow. However, we infer that gene flow between diverged canid lineages was common and that wolf-like canid relationships can be modeled

without invoking this unknown lineage. Our admixture graph analyses do uncover previously described gene flow episodes in eastern coyotes,³⁷ red wolves,³⁶ golden wolves,¹³ and dholes,³⁵ underscoring post-divergence gene flow as a major evolutionary force and the potential for hybrid speciation, as also uncovered in the Columbian mammoth (*Mammuthus columbi*), megafauna that roamed alongside dire wolves.^{39,40}

An iterative approach to nuclear genome inference

We demonstrated that iterative mapping combined with conservative consensus calling can help mitigate the effects of reference bias and DNA damage in paleogenomics. While EAGER⁴¹ uses a similar conservative approach, it employs only a single mapping round and thus captures fewer reads. Iterative assembly tools are typically limited to mitochondrial genomes (e.g., MIA⁴²) due to computational constraints,⁴³ though Meyer et al.²⁸ successfully applied two rounds of iterative mapping to the Denisovan genome (albeit with human sequence filling low-coverage gaps, introducing bias). Given the improvements we observed with multiple rounds, we recommend this approach, particularly when no close reference genome is available.

A key concern is whether iterative mapping could exacerbate rather than reduce reference bias if the seed reference is too divergent. For example, this approach might allow only the most similar ancient reads to map, masking informative variation. Our *D*-statistics^{28,44} and IBS analyses using an independent dire wolf individual (DireSP) revealed no compounding bias (Figure 2). Instead, we observed reduced bias toward the Greenland wolf seed reference and improved mapping across all read lengths, although residual bias persists for shorter reads as expected.^{30,45}

Pangenome approaches offer a strategy complementary to iterative mapping by mapping to multiple divergent species simultaneously.^{46,47} Pangenomes capture highly divergent reads and structural variants missed by single-reference mapping.⁴⁸ However, even pangenomes cannot capture haplotypes absent from their represented paths. Future optimizations combining iterative mapping with pangenome approaches will further improve paleogenome accuracy from short-read data.

Conclusion

These data offer new insights into the evolutionary history of this iconic species. The reconstructed reference paleogenome provides a resource for the broader research community and a roadmap for future studies aiming to leverage high-coverage ancient DNA. As the complex bloodlines of great houses revealed through careful genealogical study, the dire wolf's true ancestry—a hybrid lineage born from the union of two ancient canid bloodlines—demonstrates that evolution's greatest forces often arise from unexpected alliances forged in deep time. This work furthers our understanding of the dire wolf and their place in the broader patterns of canid evolution, proving that even in extinction, these magnificent beasts continue to reveal the intricate tapestry of life's history.

Limitations of the study

Our study has several limitations not discussed above. First, we had a limited number ($n = 2$) of high-quality dire wolf specimens.

While these data captured variation over a large timescale (~60,000 years), additional high-coverage samples could provide additional support for downstream analyses. Second, missing data in the canid fossil record limited our ability to resolve key events in dire wolf evolutionary history, such as divergence time estimates, and to define admixed lineages. Future discovery and sequencing of fossil specimens will be critical to refining these inferences. Lastly, there is currently no high-quality reference genome for any jackal species, limiting representation of this clade in our candid pangenome. Generating these resources will benefit future studies aiming to resolve canid evolutionary relationships.

RESOURCE AVAILABILITY

Lead contact

Requests for further information should be directed to the lead contact, Dr. Beth Shapiro (beth@colossal.com).

Materials availability

This study did not generate new unique reagents.

Data and code availability

The sequencing data underlying this study are available in the NCBI Sequence Read Archive and can be accessed with BioProject: PRJNA1222369. We also provide a gzip-compressed TSV of dire wolf-specific variant calls and genotypes with *Snpeff* annotations. These data and all supplemental tables are available in a Zenodo repository associated with this study: <https://zenodo.org/records/18624787>.

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AUTHOR CONTRIBUTIONS

Conceptualization, G.L.G., K.M.P., J.O., M.C.-J., B.L.C., and B.S.; data curation, G.L.G., K.M.P., J.O., and B.L.C.; formal analysis, G.L.G., K.M.P., J.O., C.H., M.C.-J., A.G.L., and W.T.; funding acquisition, B.L.; investigation, S.B., J.C.C., J.M., B.R.P., P.G.S.G., S.J.H., S.S., W.S., R.J.O., and W.T.; methodology, G.L.G., K.M.P., J.O., C.E.M., M.C.-J., B.L.C., and B.S.; project administration, G.L.G. and K.M.P.; resources, N.A., S.B., B.R.P., J.M., S.J.H., R.J.O., S.S., W.S., C.E.M., J.C.C., L.K., T.M., and B.S.; software, N.A., G.L.G., K.M.P., J.O., and M.C.-J.; supervision, B.L.C. and B.S.; validation, G.L.G., K.M.P., and J.O.; visualization, G.L.G., K.M.P., J.O., M.C.-J., A.G.L., and C.H.; writing – original draft, K.M.P. and G.L.G.; writing – review & editing, G.L.G., K.M.P., J.O., M.C.-J., J.C.C., N.A., W.T., O.F., S.B., T.M., L.K., J.K., A.D., M.C., J.B.P., S.O., R.J.O., S.J.H., P.G.S.G., S.S., W.S., J.M., B.R.P., G.L., A.P., S.G., A.T.L., M.-H.S.S., M.T.P.G., B.v., E.K.K., C.E.M., L.D., B.L.C., G.R.R.M., G.C., and B.S.

DECLARATION OF INTERESTS

Authors affiliated with Colossal Biosciences and/or Form Bio may hold stock and/or stock options in these companies. E.K.K., C.E.M., B.v., R.J.O., M.T.P.G., and L.D. are members of the scientific advisory board at Colossal

Biosciences. M.-H.S.S. is a consultant of Colossal Biosciences. G.R.R.M. is an investor and cultural advisor for Colossal Biosciences. G.C. is a co-founder of Colossal Biosciences and others; see arep.med.harvard.edu.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Biological samples		
Dire wolf (<i>Aenocyon dirus</i>) incisor	Sheriden Pit; Cincinnati Museum Center	DireSP; ACAD1735; VP1737
Dire wolf (<i>Aenocyon dirus</i>) cranium	Gigantobison Bay; Idaho Museum of Natural History	DireGB; ACAD18742; IMNH 48001/52
Red fox (<i>Vulpes vulpes</i>) blood samples	Colossal Biosciences Inc.	23VV001; 23VV002,23VV003, 23VV004
Deposited data		
DireSP raw sequence data	https://www.ncbi.nlm.nih.gov/biosample/46779696	SAMN46779696
DireGB raw sequence data	https://www.ncbi.nlm.nih.gov/biosample/46779697	SAMN46779697
Red Fox 1 raw sequence data	https://www.ncbi.nlm.nih.gov/biosample/47229186	SAMN47229186
Red Fox 2 raw sequence data	https://www.ncbi.nlm.nih.gov/biosample/47229187	SAMN47229187
Red Fox 3 raw sequence data	https://www.ncbi.nlm.nih.gov/biosample/47229188	SAMN47229188
Red Fox 4 raw sequence data	https://www.ncbi.nlm.nih.gov/biosample/47229189	SAMN47229189
Software and algorithms		
Fastp	Chen et al. ⁴⁹	https://github.com/opengene/fastp
Kraken2	Wood et al. ⁵⁰	https://github.com/DerrickWood/kraken2
KrakenTools	Lu et al. ⁵¹	https://github.com/jenniferlu717/KrakenTools
BWA	Li et al. ⁵²	https://github.com/lh3/BWA
PALEOMIX	Schubert et al. ⁵³	https://paleomix.readthedocs.io/en/stable/
mapDamage	Jónsson et al. ⁵⁴	https://ginolhac.github.io/mapDamage/
samtools	Danecek et al. ⁵⁵	https://github.com/samtools/samtools
DamageProfiler	Neukamm et al. ⁵⁶	https://github.com/Integrative-Transcriptomics/DamageProfiler
Picard	Auwerda et al. ⁵⁷	https://broadinstitute.github.io/picard/
Mosdepth	Pedersen et al. ⁵⁸	https://github.com/brentp/mosdepth
QualiMap	Okonechnikov et al. ⁵⁹	http://qualimap.conesalab.org/
Minigraph-Cactus	Hickey et al. ⁶⁰	https://github.com/ComparativeGenomicsToolkit/cactus/blob/master/doc/pangenome.md
Pilon	Walker et al. ⁶¹	https://github.com/broadinstitute/pilon
MIA	Hahn et al. ²⁹	https://github.com/mpieva/mapping-iterative-assembler
ANGSD	Korneliussen et al. ⁶²	https://github.com/angsd/angsd
Consensify	Barlow et al. ³⁸	https://github.com/jlapajmans/Consensify
GATK4	Auwerda et al. ⁵⁷	https://github.com/broadinstitute/gatk
bcftools	Danecek et al. ⁵⁵	https://github.com/samtools/bcftools
FLAG	Troy et al. ⁶³	https://github.com/formbio/FLAG
BUSCO	Tegenfeldt et al. ⁶⁴	https://github.com/metashot/busco
Minimap2	Li et al. ⁶⁵	https://github.com/lh3/minimap2
rotate	Durbin et al. ⁶⁶	https://github.com/richarddurbin/rotate

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
MUSCLE	Edgar et al. ⁶⁷	https://github.com/rcedgar/muscle
RAxML	Stamatakis et al. ⁶⁸	https://github.com/amkozlov/raxml-ng
MAFFT	Katoh et al. ⁶⁹	https://github.com/GSLBiotech/mafft
IQ-TREE2	Nguyen et al. ⁷⁰	https://github.com/iqtree/iqtree2
ASTRAL II	Mirarab et al. ⁷¹	https://github.com/smirarab/ASTRAL
PAML	Yang et al. ⁷²	https://github.com/abacus-gene/paml
DeepVariant	Yun et al. ⁷³	https://github.com/google/deepvariant
mash	Ondov et al. ³³	https://github.com/marbl/Mash
SequenceTools	Schiffels et al. ⁷⁴	https://github.com/stschiff/sequencetools
admixtools2	Maier et al. ³⁴	https://uqrmaie1.github.io/admixtools/articles/admixtools.html
Other		
Dire wolf iterative assembly, species-specific variant calls and genotypes with SnpEff annotations.	This study	https://zenodo.org/records/18624787

METHOD DETAILS

Published ancient DNA sequencing data

We downloaded ancient DNA sequencing reads from dire wolf individuals previously sampled, including the Sheriden Pit individual (DireSP, BioSample accession SAMEA5559854) and the Gigantobison Bay individual (DireGB, BioSample accession SAMEA5559855) and three other individuals from American Falls Reservoir, Idaho (DireAFR, BioSample accession SAMEA5559852), Natural Trap Cave, Wyoming (DireNTC, BioSample accession SAMEA5559856), and Guy Wilson Cave, Tennessee (DireGWC, BioSample accession SAMEA5559853) publicly released under BioProject accession PRJEB31639 (Table S1).

Provenance and sampling of dire wolf individuals

We obtained permission to collect new samples from two dire wolf individuals that previously yielded authentic dire wolf mitochondrial and nuclear DNA.¹¹ The first dire wolf, referred to as “DireSP” (Cincinnati Museum Center; VP1737; ACAD1735; DireSP), is from Sheriden Pit site in Indian Trail Caverns, Ohio, and an incisor tooth previously sampled at the root (BioSample accession SAMEA5559854). We sampled an additional incisor from this individual for the present study. The second dire wolf, referred to as “DireGB” (Idaho Museum of Natural History; IMNH 48001/52; ACAD18742; DireGB) is from Gigantobison Bay site at American Falls Reservoir, Idaho, and presents a complete cranium with intact teeth. The left petrosal was previously sampled (BioSample accession SAMEA5559855) and the right petrosal was sampled for this study. The Sheriden Pit individual could have been deposited before 13,000 years ago, based on calibrated radiocarbon dates from other bone material derived from the same layer,²³ but has not been radiometrically dated, and the Gigantobison Bay individual is probably older than 72,000 years, based on the age of volcanic material from a nearby stratum.²⁴

Sample preparation and ancient DNA extraction

For the Sheriden Pit individual (BioSample accession SAMN46779696, we extracted ancient DNA from an incisor (Cincinnati Museum Center VP1737). We used a Dremel with drill attachment to obtain bone powder from the interior of the root. We rinsed the obtained bone powder in a 0.5% bleach solution for fifteen minutes followed by three water washes before proceeding with extraction. For the Gigantobison Bay individual (BioSample accession SAMN46779697), we extracted ancient DNA from the incus bone recovered from the right petrosal of a complete skull (IMNH 48001/52). Ossicles are an optimal source of DNA preservation.^{75,76} We rinsed the whole incus in a 0.5% bleach solution, followed by three water washes. We extracted DNA for both individuals using the Buffer D option, a protocol optimized for recovering short, degraded molecules.⁷⁷ For the Sheriden Pit sample, we extracted DNA from two 5 mg aliquots of bone powder following bleach pre-treatment. From these two DNA extracts we generated three single stranded libraries each.⁷⁸ Following the initial round of extractions we observed a non-dimineralized bone powder pellet in one of the two lysate tubes and performed a re-digestion of this bone powder following Rohland et al. (2018).⁷⁷ From the DNA isolated after this re-digestion we generated two more single-stranded libraries.⁷⁸ For the Gigantobison Bay individual, we extracted DNA from the whole incus, which was completely dissolved following overnight lysis. We then prepared single-stranded libraries from the extracts as per Kapp et al. (2021).⁷⁸

We generated a single extract from the Gigantobison Bay ossicle, from which we made six libraries. All pre-PCR work, including sample preparation, DNA extraction, library construction, and quality control sequencing, was carried out in a dedicated ancient DNA lab at the University of California, Santa Cruz.

Massively parallel paired-end short DNA read sequencing

We prepared a total of nine Illumina libraries from the sample of the Sheridan Pit dire wolf (DireSP) and six Illumina libraries from the sample of the Gigantobison Bay dire wolf (DireGB). We sequenced all libraries on a Nova-seq 6000 using version 1.5 chemistry, loading 0.3nM final library concentration. For the DireSP sample, we obtained 1,320gb (R1+R2) of raw sequencing data or 13.2 billion total paired end reads (R1+R2). For the DireGB sample, we obtained 7,814gb (R1+R2) or 78 billion total paired end reads (R1+R2). We report the total number of raw sequencing reads generated per run, library, and individual in [Table S1](#). For subsequent steps of parallelized processing, we split sequencing reads into batches of 50,000,000 reads.

Sequence read pre-processing and quality control

We trimmed raw sequencing reads using *fastp*⁴⁹ (i) to remove adapter sequence, low quality ends of reads, and reads less than 35 nucleotides, (ii) merge and correct overlapping reads with a minimum of 15 nucleotide overlap, and no more than three differences, and (iii) trimmed homopolymer G and X sequences at the 3' end (-q 25 -L 35 -m --overlap_len_require 15 --overlap_diff_limit 3 -c -g -x). We classified sequence contaminants using *Kraken 2*⁵⁰ and a confidence of 0.8 against the pre-compiled 8 GB database MiniKraken DB_8GB constructed from complete bacterial, archaeal, and viral genomes in RefSeq (2017-10-18). We used *combine_kreports.py* from *Kraken Tools*⁵¹ to merge all metagenomic classification reports generated for split read batches. We calculated endogenous content from initial sequencing runs as total unique reads mapped/total unique reads sequenced post-quality control for each specimen. We estimated endogenous content for DireSP to be 5.20% and DireGB to be 6.95%. Given the higher endogenous content and lower duplication rates for DireGB libraries ([Table S1](#)), we chose to sequence the DireGB libraries to saturation. We report full statistics on sequencing reads before and after processing for both published and novel ancient DNA sequencing libraries in [Table S1](#).

Reference genome assemblies and annotations

We selected the Greenland gray wolf (*Canis lupus*) chromosome-level (including Y) genome assembly (GenBank GCA_905319855.2)²⁵ as our primary single-reference genome assembly of a modern canid species, and utilized additional canid genome assemblies^{16,79–95} throughout this study as referenced in [Table S2](#). We surmised that the Greenland gray wolf would serve as a suitable scaffold reference due to the geographic isolation of Greenland wolves relative to other gray wolves,⁹⁶ the fully assembled chromosomes including Y and high contig N50 relative to other canid genome assemblies ([Table S2](#)), and to previously inferred phylogenies that suggest it is no more distantly related to dire wolves than any other extant members of the Canina subtribe.

Read mapping to single reference genomes

We utilized two primary mapping strategies for the majority of the analyses in this study. Strategy 1 was a CPU-based accelerated Sentieon implementation of the *BWA* 'aln' algorithm⁵² and seed lengths of '-l 16,500' nucleotides, maximum edit distances set by a fraction '-n 0.01' of missing alignments, and '-o 2' maximum gap opens.⁹⁷ Strategy 2 was a GPU-accelerated NVIDIA Clara Parabricks 'fq2bam' implementation of the *BWA* 'mem' algorithm with settings for short reads ('-M -k 19 -r 2.5').

Similarly, when mapping ancient reads, we used two strategies for removing duplicate reads and assessing DNA damage. First in the "standard" strategy, we used the *PALEOMIX*⁵³ 'rmdup_collapsed' function and duplicate removal using *samtools*⁵⁵ 'rmdup'. We performed base recalibration with *mapDamage*⁵⁴ to rescale base quality scores in consideration of 12 nucleotides from both 5' and 3' ends of sequencing reads ('-rescale-length-5p 12' and '-rescale-length-3p 12'). We profiled Ancient DNA damage patterns using *DamageProfiler*.⁵⁶ The second strategy was "stringent," where we merged each sequencing run across libraries and marked duplicates again using 'MarkDuplicates' from the Picard toolkit⁵⁷ to remove any remaining duplicate reads at the library level. After additional duplicate removal, we ran *mapDamage* again to assess overall DNA composition, read length distributions, fragmentation and misincorporation rates and C-to-T and G-to-A transition frequencies at relative positives across reads.

We assessed alignment statistics using *samtools*⁵⁵ 'idxstats' and 'flagstat', *mosdepth*,⁵⁸ and *QualiMap*⁵⁹ 'bamqc' after removal of marked duplicates: (i) *depth of coverage* as the # of unique reads mapping across all reference bases reported as a histogram and the mean # of unique reads mapping, (ii) *breadth of coverage* as % bases of reference genome covered by given #s of unique reads reported as a histogram and the % reference at 5× coverage by unique reads, and (iii) *physical coverage* or cumulative length of post-filtering and metagenomic classification of collapsed reads expressed as a multiple of the expected 2.8B nucleotides composing the total canid genome size reported from flow cytometry in coyotes, gray wolves, and domestic dogs.^{98,99} As *QualiMap* does not exclude low mapping quality reads, unlike *mosdepth*⁵⁸, alignment statistics may be overestimated.

Sex inference of ancient specimens

We found support for both DireGB and DireSP individuals as male after mapping libraries to the Greenland gray wolf reference using *BWA* 'aln' with stringent duplication removal (see *Read Mapping 1B*). We calculated the Rx ratios^{26,27} between chromosome X depth and autosomal mean depth of coverage and 95% confidence interval (CI) lower and upper bounds for both individuals, and

designated a 95% CI upper bound under 0.6 as male and a 95% CI lower bound over 0.8 as female.^{26,27} The 95% CIs for the ratios of reads aligning to chromosome X versus autosomes placed between 0.539 and 0.554 for DireSP and 0.538 to 0.554 for DireGB.

While the ratios of reads mapping to the reference gray wolf X chromosome ($1.0\times$ for SP and $5.7\times$ for GB) and autosomes ($2\times$ for SP and $10\times$ for GB) supports these individuals as male, we did not observe similar coverage over the reference gray wolf Y chromosome ($0.1\times$ for SP and $0.4\times$ for GB), which may be unsurprising: Y chromosomes are notoriously difficult to assemble given highly repetitive regions and high homology to the X chromosome, and is roughly 1/20th the size of the X chromosome, so mapping sequencing reads from ancient DNA to the Y chromosome may be less stable than mapping to the X chromosome. Therefore, we took a closer look at reads mapping to Y-encoded genes as a secondary confirmation.

We examined depth of coverage by dire wolf reads at the Y-encoded gene *TSPY* because it exhibits relatively high copy number variation among modern gray wolves¹⁰⁰ and has high copy numbers in both humans and cattle.^{101,102} Thus, for males we expect to see a relatively higher abundance of reads aligning to *TSPY* than other windows of the assembled Y chromosome sequence. We identified coordinates for the *TSPY* gene annotated on the Greenland gray wolf (*C. lupus*) assembly (GCA_905319855.2) at chrY:594247–596596 (ENSCAFG00865029519 in our gene annotation). We then calculated the depth of coverage by collapsed dire wolf reads aligned using *bedtools*¹⁰³ and *mosdepth*⁵⁸ across *TSPY* and 10,000 randomly sampled 2,349 nucleotide windows from assembled chromosome Y and 10,000 windows randomly sampled from any assembled autosome. We observed a high depth of coverage over *TSPY* ($0.20\times$ in DireSP and $0.72\times$ in DireGB), relative to depths of coverage from equally-sized 2,349 nucleotide windows randomly sampled 10,000 times over the assembled gray wolf Y chromosome ($0.103\times \pm \text{SD: } 0.397\times$ in DireSP and $0.362\times \pm \text{SD: } 1.05\times$ in DireGB), which, while lower than read depth over randomly sampled windows across autosomes ($1.87\times \pm \text{SD: } 0.578\times$ in DireSP and $10.6\times \pm \text{SD: } 2.86\times$ in DireGB), supported the presence of male sex chromosomes.

Multi-canid pangenome construction and read mapping

Ancient DNA mapped to multi-species pangenomes is expected to reduce reference bias from mapping to a single, divergent ref.⁴⁸ Thus we generated a multi-species pangenome of canids using the graph-based *Minigraph-Cactus* (MC) pipeline,⁶⁰ which starts from a reference assembly and iteratively introduces structural variation from aligned genome assemblies and, by excluding centromeric and highly repetitive sequences, can be more efficient for mapping sequencing read data.¹⁰⁴ We constructed the graph using the Greenland gray wolf (GenBank GCA_905319855.2) as our initial reference genome to be consistent with the coordinate system of our single reference alignments, and incorporated sequence variation from 6 chromosome-level assembled genomes that had the highest contig N50 and most complete set of chromosomes from two *Canis* spp. (domestic dog and dingo) and four other canid species and 3 scaffold-level assemblies from dhole (*Cuon alpinus*), bush dog (*Speothos venaticus*), and red fox (*Vulpes vulpes*) for broad phylogenetic representation of the canid lineage (Table S2).^{16,25,79–87,89–95,105} We implemented a custom script ('buildPangenome.sh') launched from the *cactus docker image* (v2.7.1) to construct a pangenome bidirectional sequence variation graph. We built a graph index and mapped unclassified, trimmed reads using *Giraffe*¹⁰⁶ included in *vg tools*.¹⁰⁷ Only reads mapped to the mitochondrial sequence (MT) were extracted and utilized for dire wolf MT reconstruction. For downstream analysis, we surjected reads aligned to the multi-canid pangenome graph to the Greenland gray wolf (canLor) reference assembly. We observed that in pangenome alignments by *Giraffe*,¹⁰⁶ short reads were occasionally aggressively clipped, resulting in a small fraction of bases within the overall read aligning. This phenomenon was most pronounced in the dire wolf individuals. We therefore filtered pangenome alignments to remove reads which had more than 20% of their bases soft-clipped, and reads which had a non-clipped length of less than 35 bases.

Paleogenome reconstruction via iterative mapping and polishing

We reconstructed the dire wolf nuclear genome by iteratively mapping and polishing to a divergent seed reference, the Greenland gray wolf (GenBank GCA_905319855.2), using ancient DNA to progressively assemble a less divergent reference sequence. We performed initial polishing from the DireGB BAM file (see Read Depth) by employing the *Pilon* polisher⁶¹ in diploid mode to correct sequences for insertions, deletions, and single nucleotide polymorphisms (SNPs) and perform local reassembly of each chromosome or contig ('-changes -fix all -diploid').

For subsequent iterations, we allowed each library from DireGB to map to the polished sequences of autosomes, sex chromosomes, and unassembled contigs from the previous iteration, excluding the mitochondrial sequence which we assembled separately. We mapped unclassified and trimmed paired-end reads, merged and unmerged, using the *BWA* 'mem' and standard deduplication with *PALEOMIX*⁶³ (see Read Mapping 2A). Unmapped reads were discarded, libraries merged, and sequence polished. We carried out this process for 6 iterations in total and evaluated alignment statistics and quality metrics for each iteration (Table S4).

In order to reduce the computational load of assembling the mitochondrial genome and reduce reference bias in mapping, we first mapped the complete libraries of DireGB to the canid *Minigraph-Cactus* pangenome to identify mitochondrial reads in our data. We surjected all reads that mapped to mitochondrial scaffolds of the canid pangenome to the Greenland gray wolf mitochondrial (MT) sequence and reverse complemented the MT sequence to orient to strand with the D-loop control region. We then filtered reads that were unmerged, unmapped, and not in primary alignment using *samtools*⁵⁵ 'view' function (-F 261) and removed duplicates with *Picard*⁵⁷ 'MarkDuplicates' function. The filtered and duplicate-removed mitochondrial BAM was the final input from which we extracted reads using *samtools*⁵⁵ 'fastq' function (-n), resulting in 29,104 reads from the DireGB libraries.

We used *MIA*,⁴² an iterative mapping approach, to assemble the dire wolf mitochondrial genome with the gray wolf genome as the seed reference (NCBI Accession NC_008092.1), giving flags (-c -C -k 14 -U) and -s with an ancient DNA substitution matrix. We filtered the consensus assembly with “relaxed”, requiring 3× coverage and 66% consensus. All sites that did not pass these filters were set to ‘N’. For all downstream analyses, we used the relaxed filtered assembly (Figure S5). We estimated a 34.5× coverage mitochondrial consensus genome for DireSP and a 90.171× coverage mitochondrial genome for DireGB, with 3.35–21.26% N content (Table S3).

We performed a final mapping of ancient DNA sequencing libraries to nuclear sequences resulting from the 6th iteration of mapping and polishing plus the iteratively assembled mitochondrial sequence. We aligned using *BWA* ‘aln’ with standard *PALEOMIX*⁵³ deduplication (Read Mapping 1A). We called pseudo-haploid consensus sequences by random read-sampling consensus and by variant-based consensus. In both cases, sequences below 5× unique read depth were rendered as hard-masked Ns, and we required a minimum base quality of BQ = 30 and mapping quality of MQ = 25. We performed random read-sampling using *ANGSD*⁶² allele counting (-doCounts 1 -minQ 30 -minMapQ 25 -dumpCounts 3) followed by *Consensify*³⁸ to call consensus from reads mapped at 5× to 30× depth and 4/5 allelic support (-min 5 -max 30 -n_matches 4 -n_random_reads 5). We called a variant-based consensus to consider read depth and quality scores at variant and invariant sites. We called genotypes using *GATK*⁵⁷ *HaplotypeCaller* emitting in gVCF mode, then converted to VCF using *bcftools*⁵⁵ ‘convert’ function, filtered variant calls (minimum QUAL = 20, BQ = 30, MQ = 25, DP = 5). In the case of overlapping alleles at variant sites, we enforced each genotype to represent homozygous genotype of the allele with highest depth, which should minimize bias from DNA damage-induced cytosine deamination producing false variant calls, using the ‘remove-overlaps’ plugin for *bcftools*.⁵⁵ For variable sites, overlapping variants with the highest QUAL values were retained, and for overlapping sites that contain invariant sites, we generated pseudo-QUAL values using the highest QUAL value of a known variant site scaled by the depth of the invariant site prior to filtering. Any sequences not covered in the resulting VCF were hard-masked as Ns in the final consensus.

Gene annotation and orthology analysis

We annotated genes using the fully automated *FLAG* tool⁶³ implemented in the ‘FLAG: Eukaryote Gene Annotation’ workflow on the Form Bio platform (formbio.com). For the reconstructed dire wolf assembly, we included fully assembled transcript data from domestic dog (GCA_000002285.4), gray wolf (GCA_905319855.2), and dingo (GCA_003254725.2) as evidence for gene prediction modules and gene symbol attribution. We performed a *BUSCO*⁶⁴ analysis using default settings and reported the total number of orthologs found in the mammalian database (*n* = 9226). We assessed mean depth of coverage by all aligned short-read libraries over all annotated genes (length-weighted average of mean depth per exon) in the reconstructed dire wolf genome sequence using *mosdepth* (Table S5).

Consensus calling from iterative reconstruction

We mapped short-read genomes of modern canid species sourced from publicly available datasets (Table S8),^{11,13–16,25,35,96,108–112} the new sequencing libraries from DireGB and DireSP individuals, and the previously published libraries of dire wolf individuals¹¹ to the reconstructed dire wolf genome reference (aenDir) using *BWA* ‘mem’ with standard deduplication with ‘Mark Duplicates’ (Read Mapping 2A).

We called consensus sequences using a random read-sampling approach. We used *ANGSD*⁶² (-doCounts 1 -dumpCounts 3) to quantify depth of sequence variants with a minimum mapping and sequence quality of 25. We used these counts to generate a consensus genome using *Consensify*³⁸ for the modern (-min 10 -max 500 -n_matches 8 -n_random_reads 10) and ancient (-min 5 -max 30 -n_matches 4 -n_random_reads 5) read sets. Note in both cases, a variable site was incorporated as a change to the reference only if 80% of alternative reads samples were in agreement. Such random sampling of ancient data also decreases the likelihood of incorporating false positive variants from DNA damage signals.³⁸

Post hoc analysis of residual reference bias

We assessed residual reference bias as a disproportionate rate of derived allele sharing with the seed reference Greenland gray wolf (*C. lupus orion* aka “canLor”) genome (GenBank GCA_905319855.2)²⁵ relative to the genome of a distinct gray wolf subspecies (*C. lupus* aka “canLup”). We employed the ABBA/BABA *D*-statistic for introgression^{28,44} to quantify derived allele sharing. We designated ancestral (A) alleles as those shared with outgroup species gray fox (*Urocyon cinereoargenteus* aka “uroCin”) and/or red fox (*Vulpes vulpes* aka “vulVul”), where for site patterns of derived (D) alleles considered concordant under the expected species tree, ‘(((canLor, canLup), aenDir), outgroup/vulVul/uroCin)’ would be ‘(((D,D),A),A)’ or ‘(((D,D),D),A)’ and discordant as ‘(((D,A),D),A)’ and ‘(((A,D),D),A)’. We assessed these site patterns using haploid calls from long sequence alignment over syntenic regions to reference genomes of outgroup canid species varying in divergence.

First, we performed a long assembly-to-reference mapping with *minimap2*⁶⁵ using the preset for up to 10% sequence divergence (-asm10), and applied the *minimap2* ‘paftools call’ function to identify aligned regions (R lines) and sequence variants (V lines) relative to outgroup assemblies from most-to-least diverged from or within the wolf-like clade: the gray fox reference assembly (GenBank GCA_032313775.1)⁹³ and the red fox reference assembly (GenBank GCA_964106925.2),⁹² the maned wolf (GenBank GCA_028533335.1, *Chrysocyon brachyurus* aka “chrBra”) reference assembly,¹⁶ and the African wild dog (*Lycaon pictus* aka “lycPic”) reference assembly (Yggdrasil DB12).⁹⁵ To those reference genomes, we aligned assembled genome sequences of

outgroups red fox and gray fox, when not used as reference; the seed-reference Greenland gray wolf (*C. lupus orion* aka canLor) and a non-reference genome assembled for a North American wolf (GenBank GCA_003160815.1, *C. lupus* aka canLup)⁹⁴; and the pseudo-haploid consensus genome sequences of dire wolf individual DireGB called from random read-sampling *ANGSD*⁶² + *Consensify*³⁸ from alignment to either the reconstructed ancient dire wolf (*Aenocyon dirus* aka “aenDir”) genome reference (“iterative”) or its seed-reference sequence of Greenland gray wolf (“divergent”).

We restricted analyses to regions over each outgroup genome aligned by gray wolf and dire wolf genomes, comprising a total of 171,599,977 gray fox aligned sequence bases, 247,756,445 maned wolf aligned bases, and 244,509,933 African wild dog aligned bases, where these taxa should have long-range synteny for sequence alignment and variant calling. We then computed global *D*-statistic as $(\# \text{ ADDA sites} - \# \text{ DADA sites}) / (\# \text{ ADDA sites} + \# \text{ DADA sites})$ over canid syntenic regions of respective reference assemblies. We computed standard errors via a block jackknife procedure²⁸ to divide the outgroup genomes into 1 megabase equally-spaced contiguous blocks, and adjusted global *D*-statistics by this standard error to obtain Z-scores (Table S6).

We mapped DireSP sequence reads using BWA mem as described above to either the iteratively reconstructed dire wolf or gray wolf reference assemblies. Using the *samtools*,⁵⁵ we sampled 500,000 reads from each alignment file and parsed their cigar strings to calculate the identity-by-state (IBS) for each read. IBS was defined as the (number of aligned bases (M) - number of mismatches (NM))/aligned sequence length (M). Reads were binned by their aligned sequence lengths and average IBS was calculated for each bin.

Mitochondrial and nuclear phylogenetic analysis

In order to place our newly assembled mitochondrial sequence in phylogenetic context, we downloaded publicly available canid mitochondrial genomes (Table S9). We used *rotate*⁶⁶ to ensure all circularized sequences started at the same position and aligned the sequences with *MUSCLE* v3.8.1.⁶⁷ We used *RAXML* v8.2.12⁶⁸ to build a maximum likelihood phylogeny (-m GTRGamma -f a -# 100 -p 12345 -x 12345) under the GTRGAMMA substitution model with 100 bootstrap replicates. The dire wolf mitochondrial genomes formed a clade outside of other wolf-like canids (Figure S5) in a topology similar to that estimated previously.^{11,13,113}

To localize the placement of the dire wolf in the broader canid phylogeny, we set out to build a nuclear phylogeny using neutral windows of genomic sequence. We defined neutral windows using *bedtools*¹⁰³ makewindows (-w 25000 -s 1025000 -i winnum), which defined uniquely named 25 Kb loci at least 1 Mb apart. Windows containing any annotated protein coding genes were removed using *bedtools*¹⁰³ intersect (-v). This resulted in 1,332 possible loci to build trees from.

To assess canid sequence divergence over these windows, we aligned 14 representative canid datasets (Table S8), along with DireGB and DireSP, against the iteratively assembled paleogenome reference and obtained consensus sequencing using the *ANGSD*⁶² + *Consensify*³⁸ approach outlined above. Sequences over the neutral windows were extracted and N content was measured using the “letterFrequency” function from the *BioStrings* package in R. Only windows that exhibited $\leq 50\%$ missingness across all species ($n = 924$) were retained for further analysis. Sequences that passed these thresholds were converted to multiple sequence alignments using *MAFFT*⁶⁹ and trees were built from each loci using *IQ-TREE*⁷⁰ v2 with 100 bootstraps each with a GTR + I + G substitution model.^{114,115} Each loci tree was merged and provided as input for *ASTRAL* II^{71,116,117} to call a final consensus species tree.

Divergence time calibration

We used *MCMCtree* v4.10.7 from the *PAML* package⁷² to perform divergence time estimation. We used an autocorrelated-rates relaxed clock model with approximate likelihood calculation and the HKY85¹¹⁸ substitution model with gamma-distributed (five categories) rate heterogeneity among sites. We used gamma priors for kappa (shape = 2, scale = 10) and alpha (1, 10) in the substitution model, and gamma-Dirichlet priors for the overall substitution rate (shape = 2, scale = 40, concentration = 1) and rate drift parameter (2, 20, 1). These priors were selected to provide diffuse yet biologically realistic constraints on the nucleotide transition/transversion ratio, among-site rate variation, overall rate of molecular evolution, and rate drift (allowing for changes in evolutionary rate from parent to offspring branches) in the autocorrelated relaxed clock model, respectively.

We used two nodes as calibration points with priors based on previous studies: the root age calibrated for gray foxes (*Urocyon* spp.) (same root calibration as Perri et al. 2021¹¹) versus other canids¹¹ and the South American canid node represented by bush dog and maned wolf.¹⁶ Lower and upper bounds for the root age were 10.3 Mya (*Metlopex macconnelli*) and 20 Mya (*Leptocyon* spp.), and 1.35–4.34 Mya for the South American canid node. For both nodes, we used log-normal distributions as priors, with 0.025 for location and scale parameters. This approximated a uniform distribution between upper and lower bounds, while softening the bounds with low-probability tails (see prior distributions of node ages in Figure S9). The MCMC analysis was run for 1,000,000 iterations, with the first 10,000 discarded as burn-in, and samples collected every 100 iterations, resulting in 10,000 samples. We created visualizations of the resulting tree with divergence time estimates using the *MCMCtreeR* package¹¹⁹ in R, showing both node age uncertainty and posterior distributions of node ages (Figure 3; Table S10).

Sequence variant calling

We detected germline variants against the reconstructed dire wolf reference (aenDir) from BAMs of 35 modern canids (Table S8) and DireGB and DireSP using a joint-calling workflow implemented by *GATK4*⁵⁷: *HaplotypeCaller* for local re-assembly of haplotypes, *GenomicsDBImport* for gVCF consolidation, and *GenotypeGVCFs* for joint variant calling. We then set as missing any genotype

which had GQ < 30 or depth less than 1/3rd or greater than 3x mean individual depth using the *bcftools*⁵⁵ ‘setGT’ plugin. We also generated a variant call set comprising transversions only by filtering with *bcftools*⁵⁵ option, ‘-i ’((REF = "A" & ALT~"[TC])" | (REF = "T" & ALT~"[AG]")) | (REF = "C" & ALT~"[AG]")) | (REF = "G" & ALT~"[TC]")) & (strlen(REF) = 1 & strlen(ALT) = 1)’, which totaled 41,774,524 SNPs.

We also called variants from the multi-canid pangenome alignments surjected to the Greenland gray wolf reference (canLor) using the GPU-optimized NVIDIA Clara Parabricks implementation of *DeepVariant*⁷³ for 23 modern canids (Table S8) and both dire wolves. *DeepVariant* gVCFs were combined using *GLNexus*¹²⁰ with the ‘DeepVariant_unfiltered’ configuration to avoid biasing allele frequencies of non-gray wolf species. We then set as missing any genotype with GQ < 10 or depth less than 1/3rd or greater than 3x mean individual depth using the *bcftools*⁵⁵ ‘setGT’ plugin and filtered any site where fewer than 10 individuals were genotyped.

Dire wolf specific variants were those variants not present in any other canid call set from the pangenome alignments and which were called homozygous alternate genotypes in DireGB and either uncalled or also 1/1 in DireSP. These had to be supported by read depths greater than 5x but under 30x, and were filtered for genotype quality (GQ = 1) and allele depth/total depth (VAF = 2). We assigned effects on annotated genes using *SnpEff*.

Sequence distance, diversity, and divergence

We estimated the sequence distance of dire wolves from extant canids using the formula: 2 x (mutations/generation) x (divergence time/generation time), assuming a mutation rate of 4 billion substitutions per generation calibrated for *Canis* species,³¹ a generation time between 5 and 8 years for black-backed jackals and wolf-like canids, respectively,³² and a divergence time of 5.8 Mya.¹¹

We assessed sequence similarity in two ways. First, we filtered the final species variant call set (described above) for homozygous transversions by species using *bcftools*⁵⁵ ‘query’ function and compared each using a custom script. Each species in a comparison had to have the same site vary in the same way to be considered a shared allele. We then subtracted the total differences from the non-N sequence length of the reconstructed dire wolf (2,310,286,013 nucleotides) and expressed a proportion of total available sequence. In parallel, we estimated sequence distance using *mash*.³³ For each consensus sequence, we used *mash* ‘sketch’ function to hash the genome into 32-mers and assessed common 32-mers in each species in a pairwise manner.

Gene flow and admixture graph modeling

We converted autosomal variants to *EIGENSTRAT* format using ‘vcf2eigenstrat’ function from *SequenceTools*⁷⁴ for both the *GATK4* jointly-called variants against the dire wolf consensus genome and *DeepVariant* variants called against the gray wolf genome from the surjected pangenome alignments. To account for differential coverage across individuals, we then made pseudohaploid genotype calls for each individual for the full set of variants using a read sampling procedure modeled after *Consensify*³⁸ that controls for biases arising from differential sequencing coverage and errors (Figures S17–S19) by randomly selecting 3 reads at each site, or two reads at positions covered by exactly two reads, and requiring at least two bases among the selected reads match. Sites with fewer than two reads were set to missing. This procedure resulted in 33.91M aenDir-mapped and 35.06M pangenome-mapped autosomal, biallelic, mappability-filtered transversions for 17 populations represented by 37 individuals. Ultimately, alignment to the reconstructed dire wolf reference increased reference bias (attraction toward DireGB) in a manner that pangenome alignment did not (Figures S16 and S19), particularly for more divergent outgroups. As admixture tests can be particularly sensitive to such biases,¹⁸ we focused pangenome-derived results for describing canid population history and gene flow, though note that patterns and ratios of admixture in the dire wolf lineage obtained from alignments to the reconstructed dire wolf genome were similar to those inferred from pangenome aligned reads (Figure S16).

We used *D*-statistics to assess allele sharing between dire wolves and extant Canini lineages, using either red foxes or gray foxes as an outgroup. *D*-statistics were calculated using *admixturetools2* v2.0.6.³⁴ We then used the *findGraphs* function from *admixturetools2*, which uses a series of heuristics to randomly permute graphs and efficiently evaluate the fit of a large number of graphs, to search for possible well-fitting admixture graph models across varying subsets of canid species. For each set of species, we performed 100 separate iterations of *findGraphs* across 0–8 admixture events, and extracted the graph with the lowest log likelihood score as the best-fitting model across each iteration. In cases where less complex graphs provided adequate fits for a given set of species (worst residual of *Z* < 3), we did not model additional admixture events.

We began with a minimal set of species representing major canid lineages: dire wolves (*n* = 2), Eurasian gray wolves (*n* = 2), coyotes (*n* = 4), dholes (*n* = 2), African wild dogs (*n* = 3), maned wolves (*n* = 2), and red foxes (*n* = 2) as an outgroup. We also explored similar models but with either gray foxes or both red and gray foxes as outgroups. We also explored graphs in which dire wolves were each modeled individually, rather than grouped into a population. Beyond this minimal species set, we also modeled a complex graph which featured African jackals (side-striped jackal and black-backed jackal), along with an additional South American canid lineage (Pampas fox), and a North American gray wolf population (Yellowstone). Finally, we assessed admixture graphs which featured only extant canid lineages and included populations with known history of admixture (golden wolves and red wolves), to assess the sensitivity of this method to detect known instances of admixture among canid lineages.

Supplemental information

On the ancestry and evolution of the extinct dire wolf

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On the ancestry and evolution of the extinct dire wolf

Supplemental Tables

Document S1: Figures S1-19

Table S1: Sequencing library statistics for all ancient samples, related to Figure 1.

Table S2: Reference assembly accessions used in study related to Figures 1, 3 and 4.

Table S3: Alignment statistics for all ancient samples, related to Figure 1.

Table S4: Iterative mapping and polishing statistics for paleogenome reconstruction, related to Figure 1 and 2.

Table S5: Gene annotation information for paleogenome reconstruction , related to Figure 1.

Table S6: Allele sharing statistics for residual reference bias assessment, related to Figure 2.

Table S7: Genetic distance metrics, related to Figure 3.

Table S8: Short read dataset accessions used in study, related to Figure 3 and 4.

Table S9: Mitochondrial genome accessions used in study, related to Figure 3.

Table S10: Age estimates for nodes in phylogenetic tree, related to Figure 3.

Table S11: Summary of admixture methods and models, related to Figure 4.

Supplemental Figures

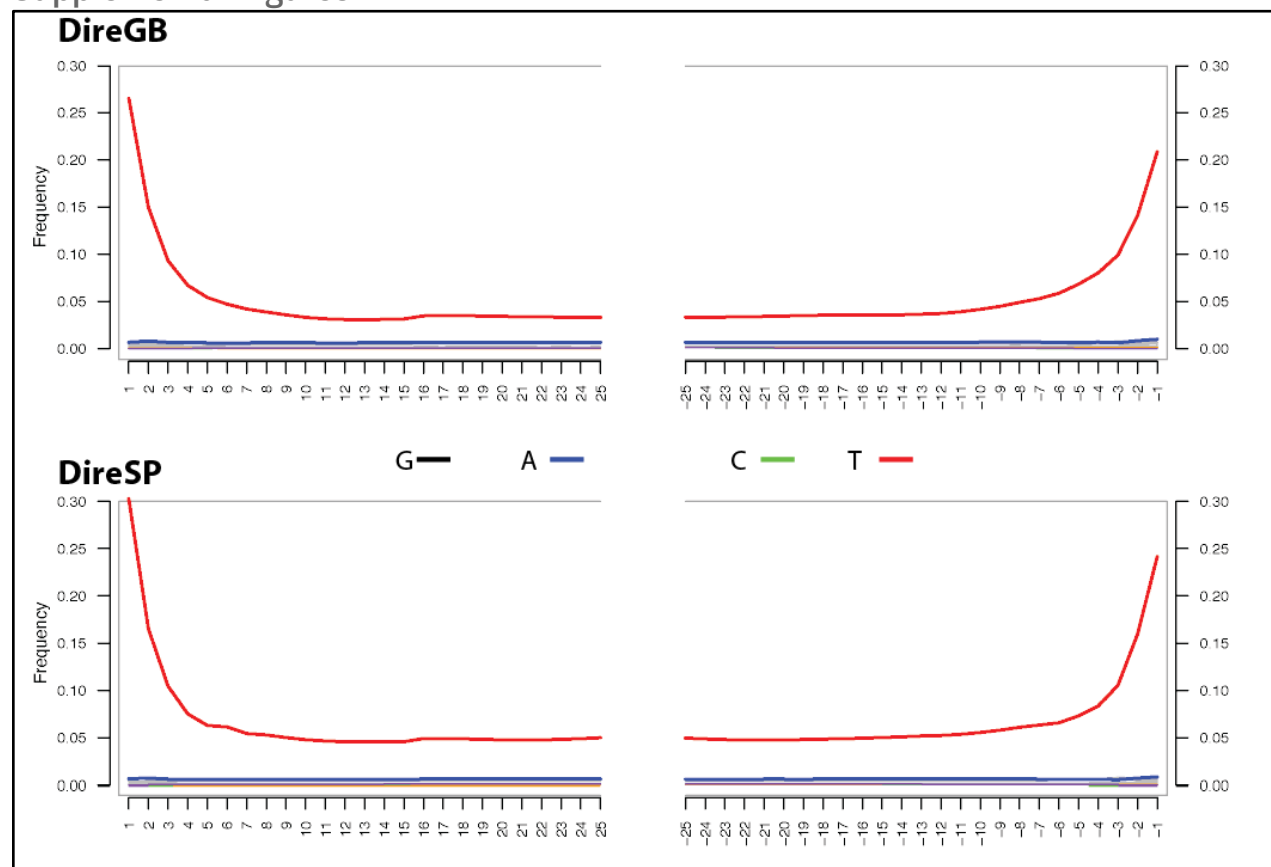


Figure S1. DNA damage profile of libraries from two *Aenocyon dirus* specimens, DireGB and DireSP, related to Figure 1. Both libraries exhibit increased cytosine deamination at both ends of molecules, typical of single-stranded libraries.

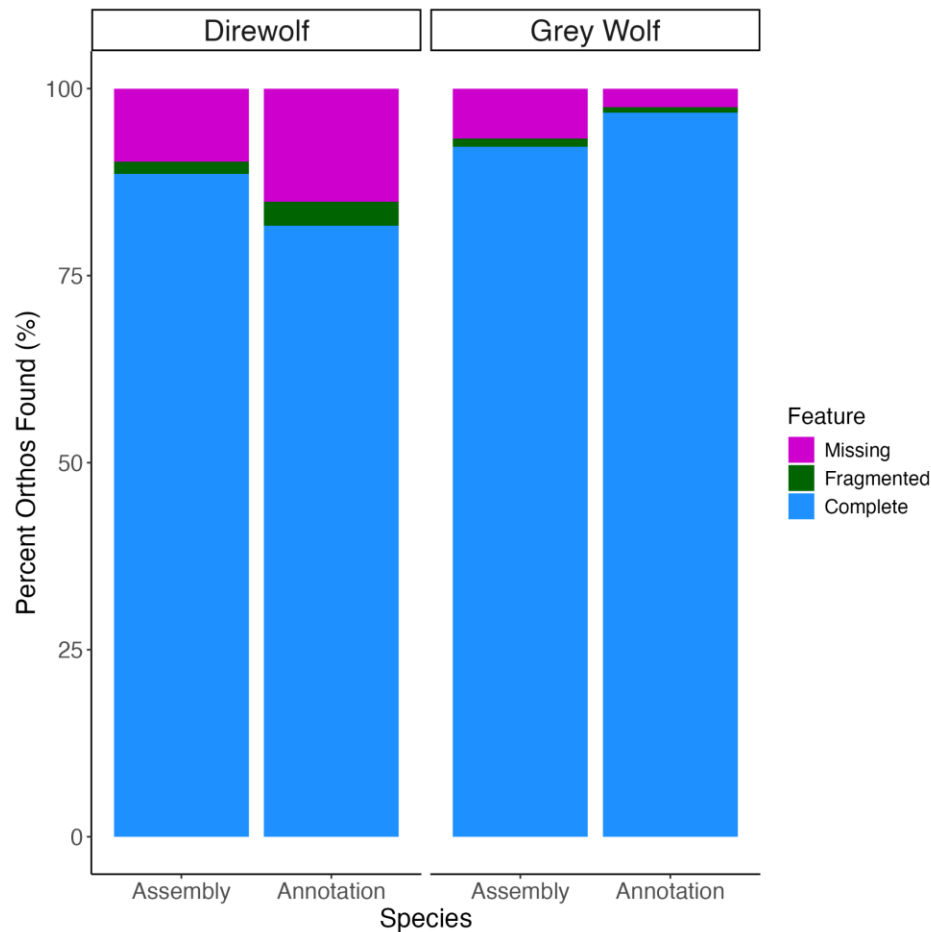


Figure S2. Completeness measures of ancient annotation, related to Figure 1. BUSCO statistics of mammalian orthologs for the dire wolf reconstructed genome assembly and annotation produced using FLAG. Grey wolf measurements reported from Ensembl. The majority of single copy protein orthologs identified in the dire wolf assembly are captured by the FLAG annotation pipeline.

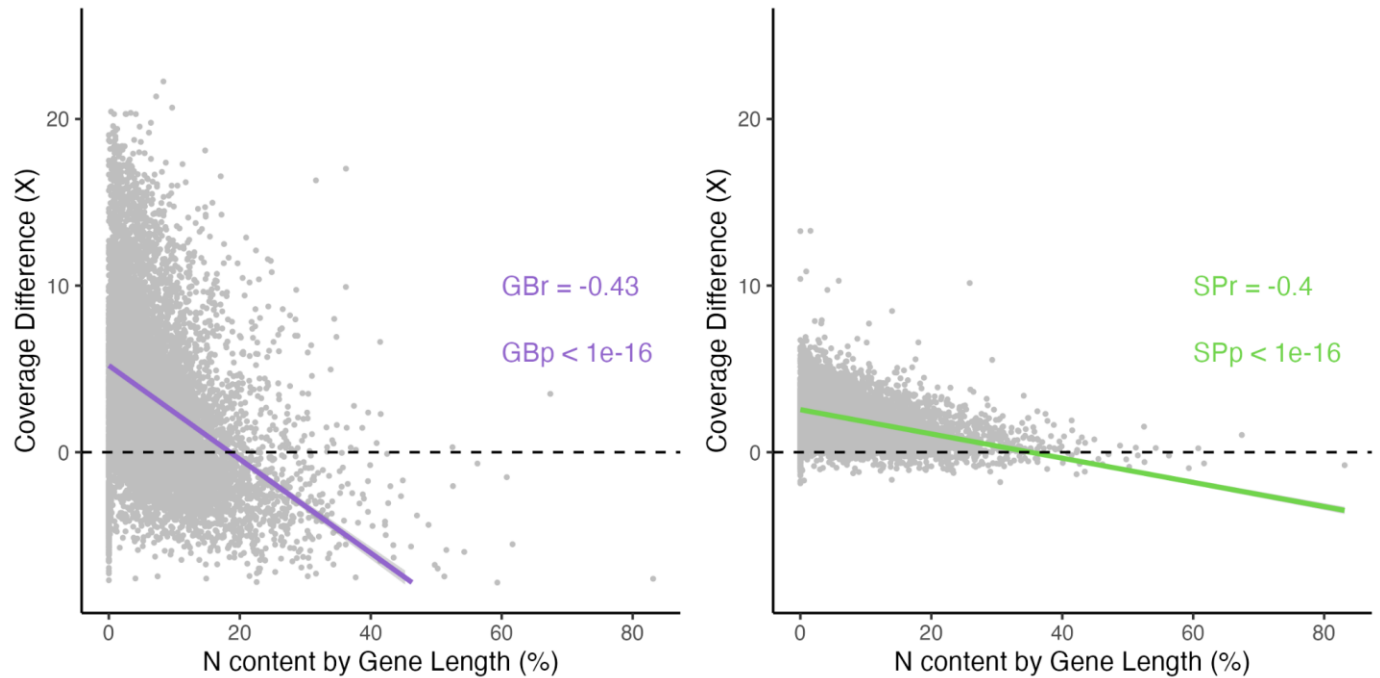


Figure S3. Decreased coverage over reconstructed dire wolf exome is correlated with missing data, related to Figure 1. The total coverage difference when mapping either DireGB (purple) or DireSP (green) to either the reconstructed dire wolf reference or the gray wolf reference. Positive values indicate improved coverage with dire wolf reference. N content is measured as the total number of N bases across the full CDS of a given gene. Only genes confidently annotated in both assemblies ($n = ?$) were used for comparison. Correlation test conducted in R.

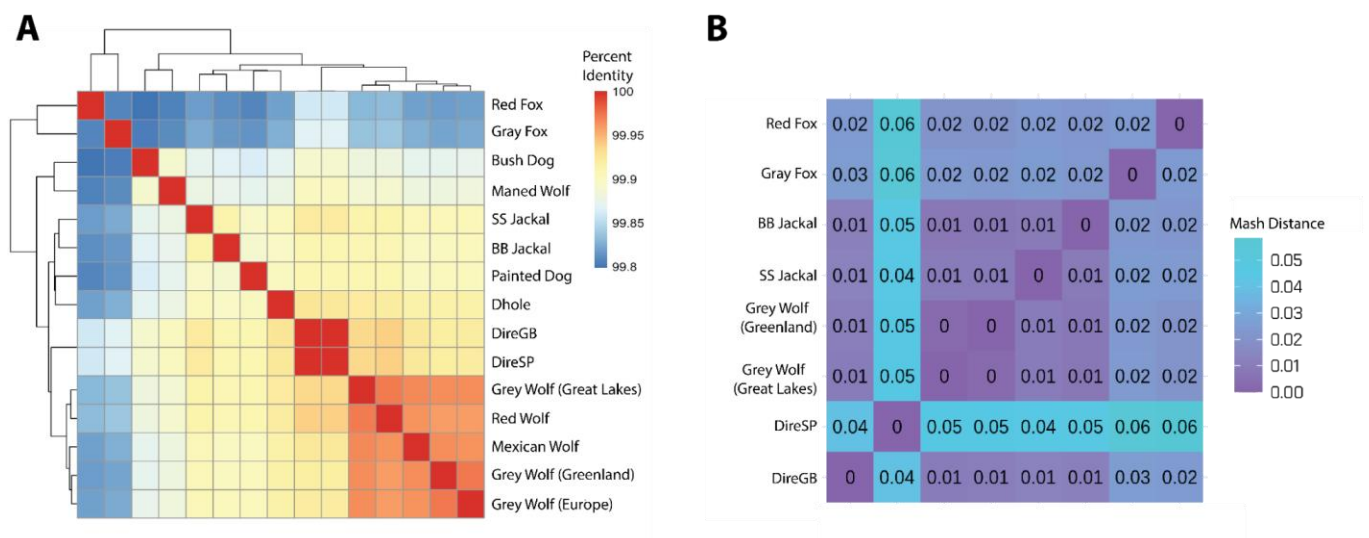


Figure S4. Sequence similarity across canid genomes, related to Figure 3. A) Heatmap showing pairwise percent similarity across a variety of canid species. Percent similarity was determined by comparing the number of shared homozygous transversions over the total non-N

proportion of dire wolf reference sequence (~2.3Gb). Warmer colors indicate greater sequence similarity (i.e. fewer shared variants). **B)** MASH distance comparing common sets of 32-mers in foxes, jackals, wolves, and dire wolves (n = 2 each). Warmer colors indicate more shared kmers and greater sequence similarity.

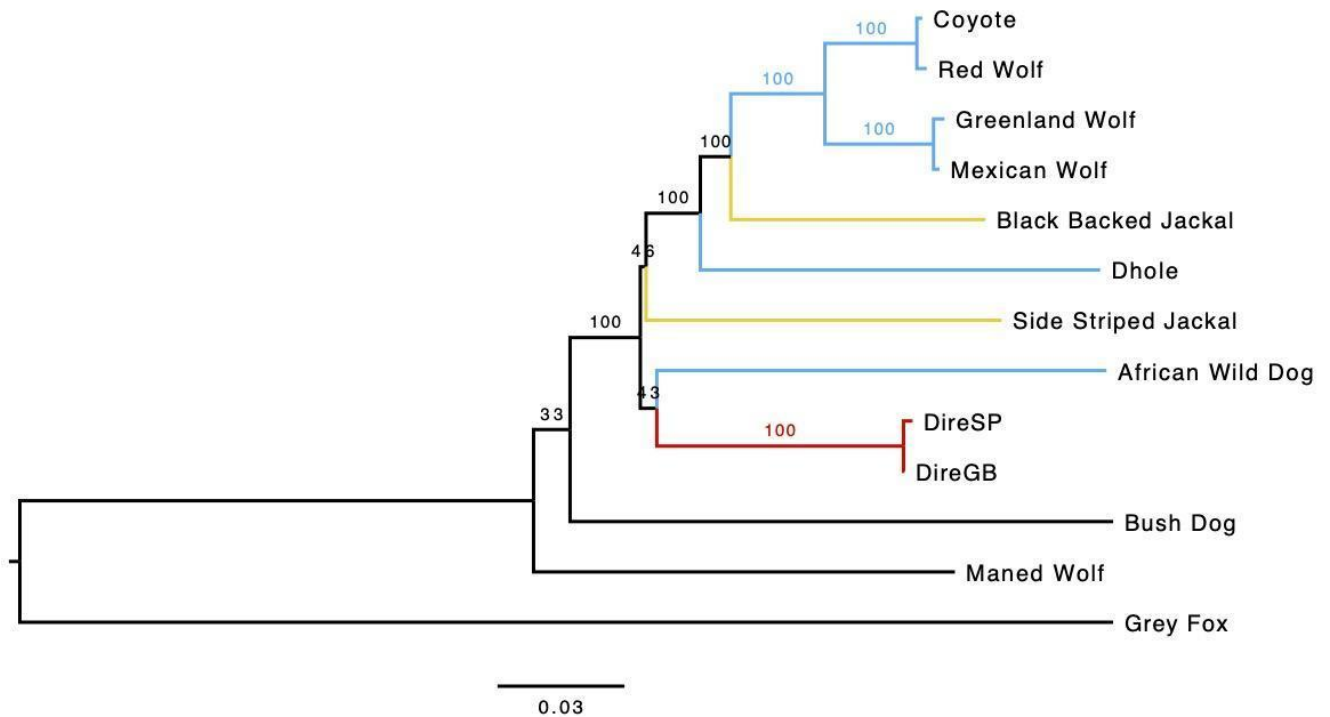


Figure S5. Phylogeny of dire wolf mitochondrial DNA, related to Figure 3. Consensus tree built from full length mitochondrial sequences from wolf-like canids (blue), jackals (yellow), dire wolves (red) and other canids (black). Bootstrap support values (n = 100 replicates) from RAxML are displayed for each node.

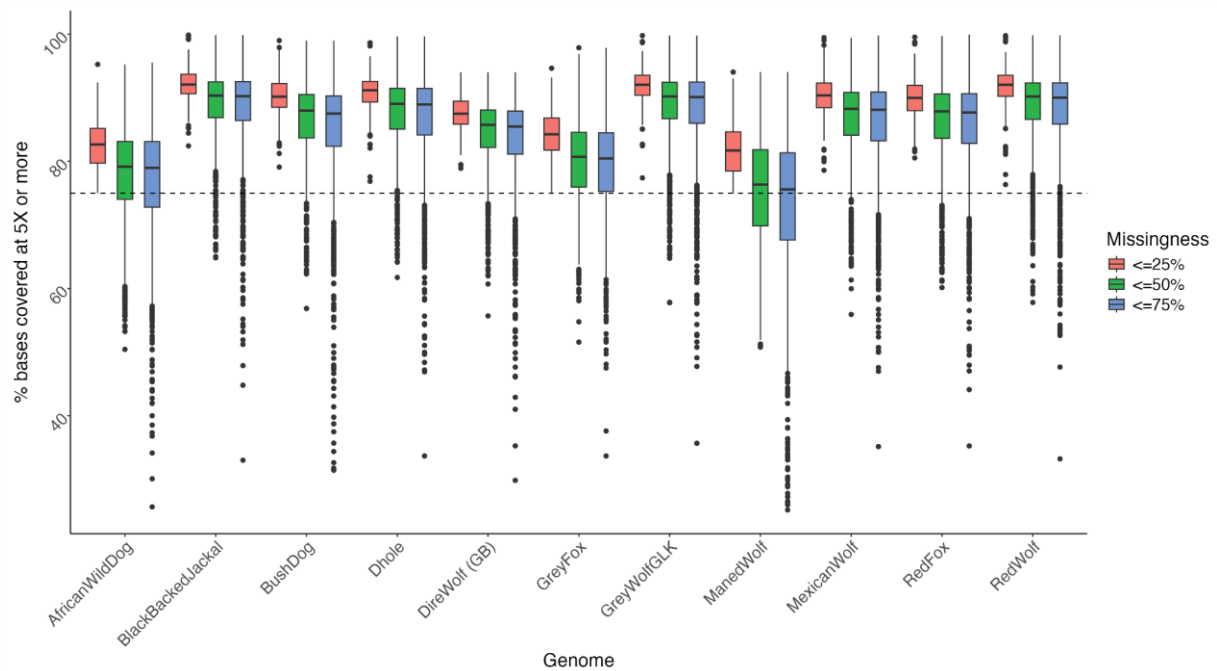


Figure S6. Missing data in phylogenetic sequence windows across species, related to Figure 3. Boxplots display distribution of 25kb neutral windows at either 25% (red), 50% (green), or 75% (blue) maximum missingness thresholds. Dotted line indicates windows with 75% of bases covered at 5X or more (i.e. 25% missingness). While additional windows are added as the missingness threshold increases, missingness does not surpass 25% on average across all thresholds and species.

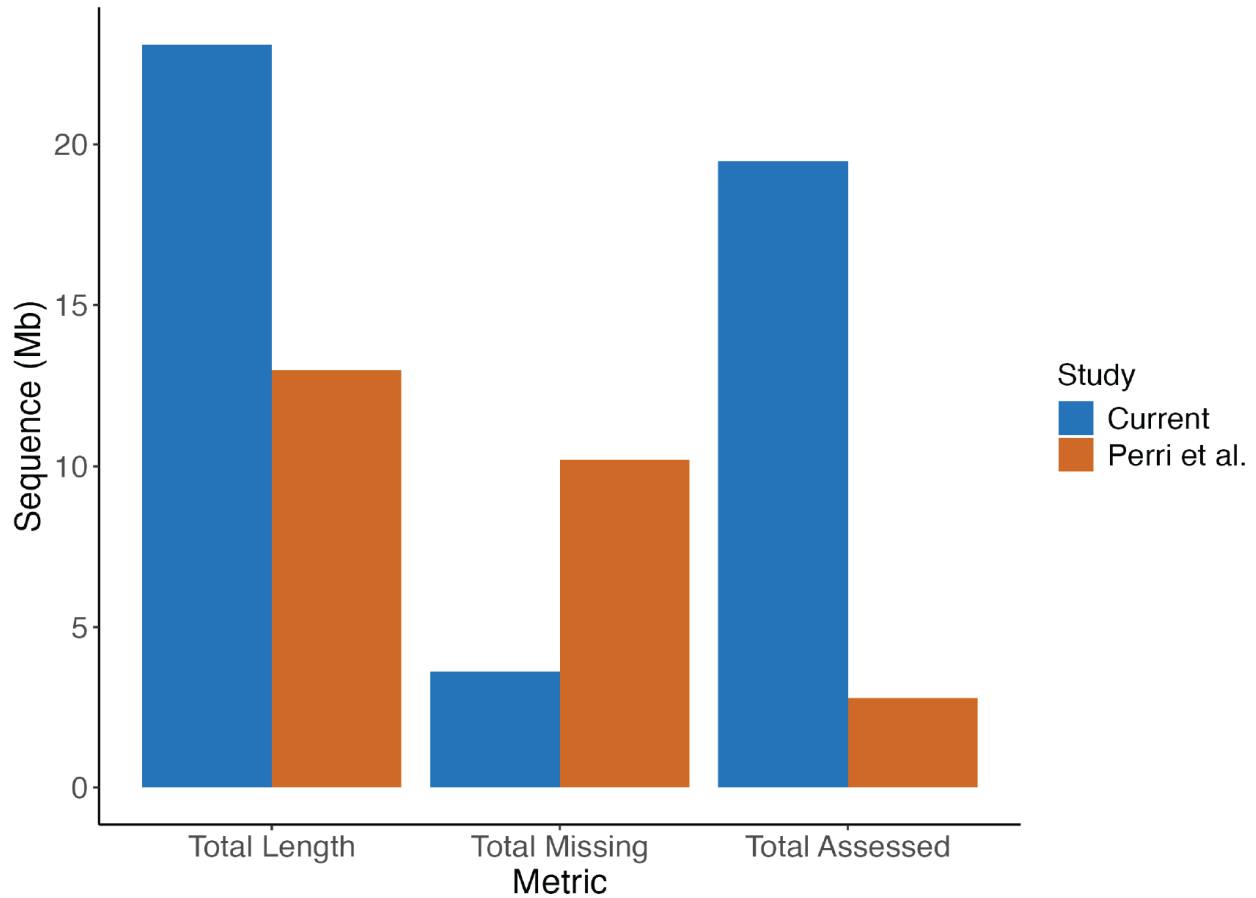


Figure S7. Alignment statistics for DireGB data utilized in nuclear phylogeny, related to Figure 3. Total aligned bases of DireGB across all windows of the alignment. For the present study, total alignment length was calculated by summing all 25kb windows used (n=924), missingness was determined by summing all N bases over these windows in DireGB, and total assessed was the difference. All values were compared to those reported in Perri et al. (2021)¹¹ for the analysis using DireGB only.

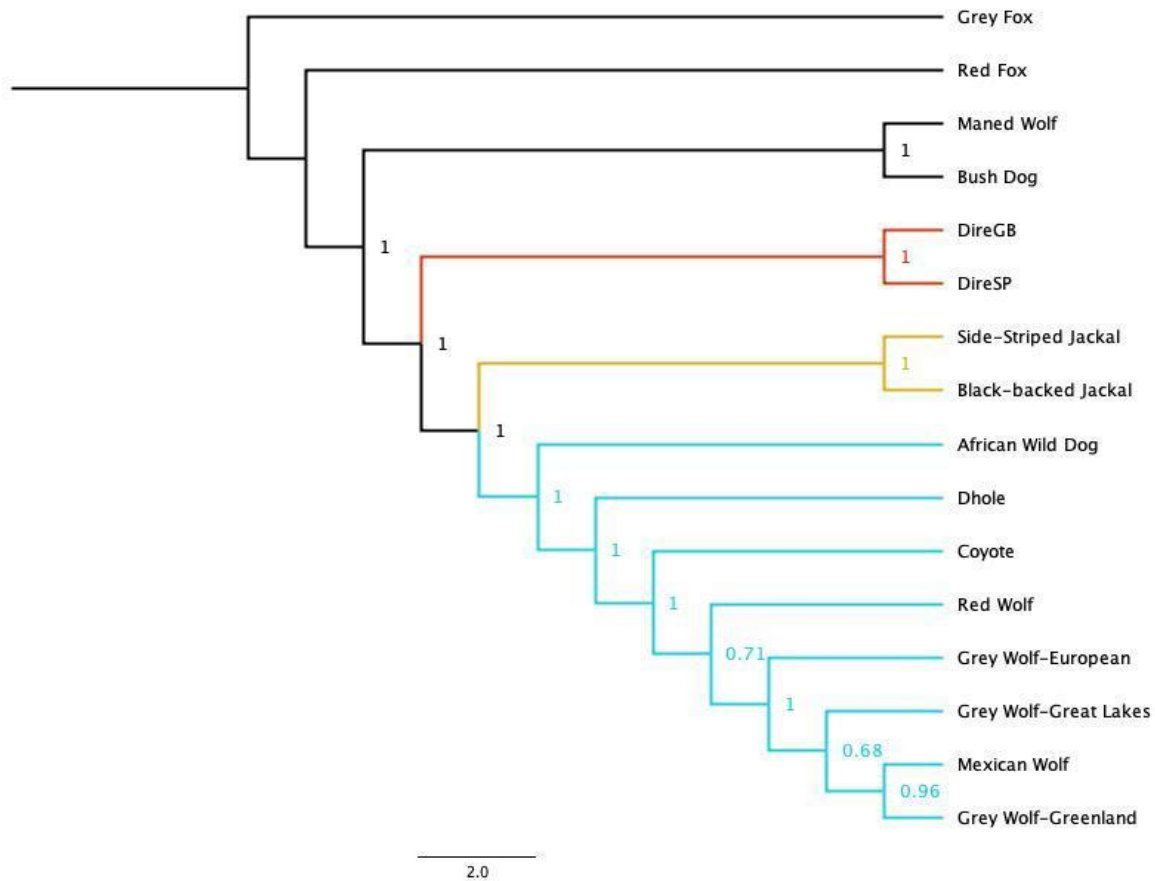


Figure S8. Nuclear phylogeny with both specimens confirms monophyletic placement of dire wolf branches, related to Figure 3. Nuclear phylogenetic relationship of dire wolves (DireGB and DireSP) from 25kb loci trees ($n = 861$) with no more than 75% missingness across all windows in all samples. Branch lengths presented in coalescence units and node labels are posterior probabilities from consensus tree calling from ASTRAL from each bootstrapped loci tree ($n=100$ replicates each). Dire wolves (red) form a monophyletic clade branching prior to the split of jackals (yellow) and wolf-like canids (blue).

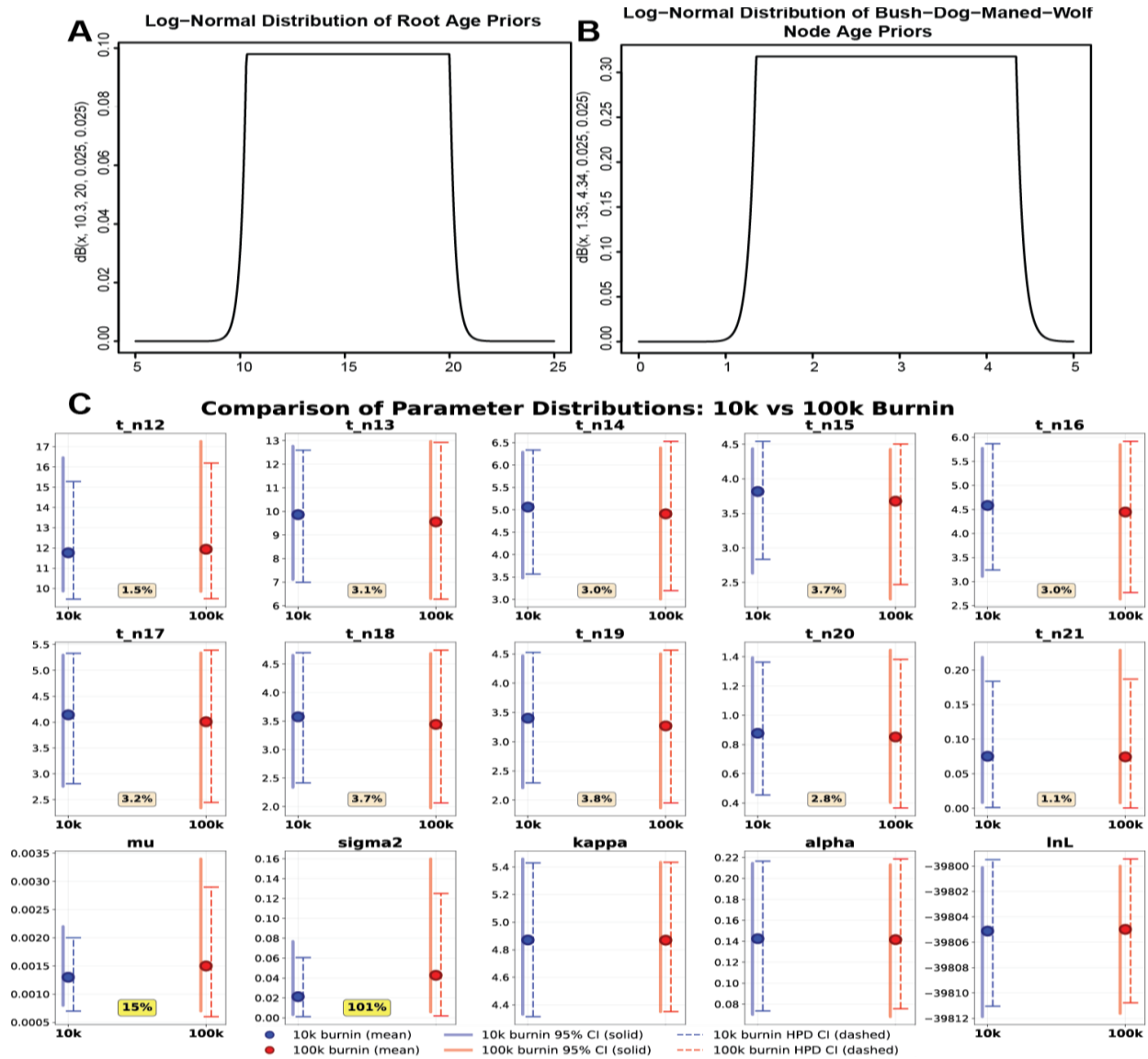


Figure S9: Log-normal distributions of age priors and posterior distribution comparisons of separate MCMCtree runs with 10,000 (10k; blue) and 100,000 (100k; red) burn-in, related to Figure 3. We used the same priors for two key nodes in the phylogenetic tree of canids: **A**) Root age prior, representing the split between gray foxes (*Urocyon* spp.) and other canids, with bounds of 10.3-20.0 million years ago (mya) based on Perri et al. (2021)¹¹; **B**) South American canid node prior, representing the divergence of bush dog and maned wolf, with bounds of 1.35-4.34 mya derived from Chavez et al. (2022)¹⁶. These priors were designed to approximate uniform distributions within specified bounds while allowing for low-probability tails beyond these limits. We tested robustness of node age estimates and parameter convergence by comparing two MCMCtree runs: **C**) MCMC chains converged even with a shorter burn-in period of 10,000 (i.e., 1% of 1-million MCMC chain) compared to 100,000. While the mean (μ) and variance (σ^2) of evolutionary rate showed large differences (yellow boxes; 15% and 101% difference in posterior mean, respectively), the effect on node age posteriors was minimal. The means of all node age estimates were within 1-3% of each other between runs (beige boxes; e.g., the difference in posterior mean for the root age, t_{n12} , was 3.2%). The difference of posterior means for the last three parameters (κ , α , $\ln L$) was in all cases less than 1%.

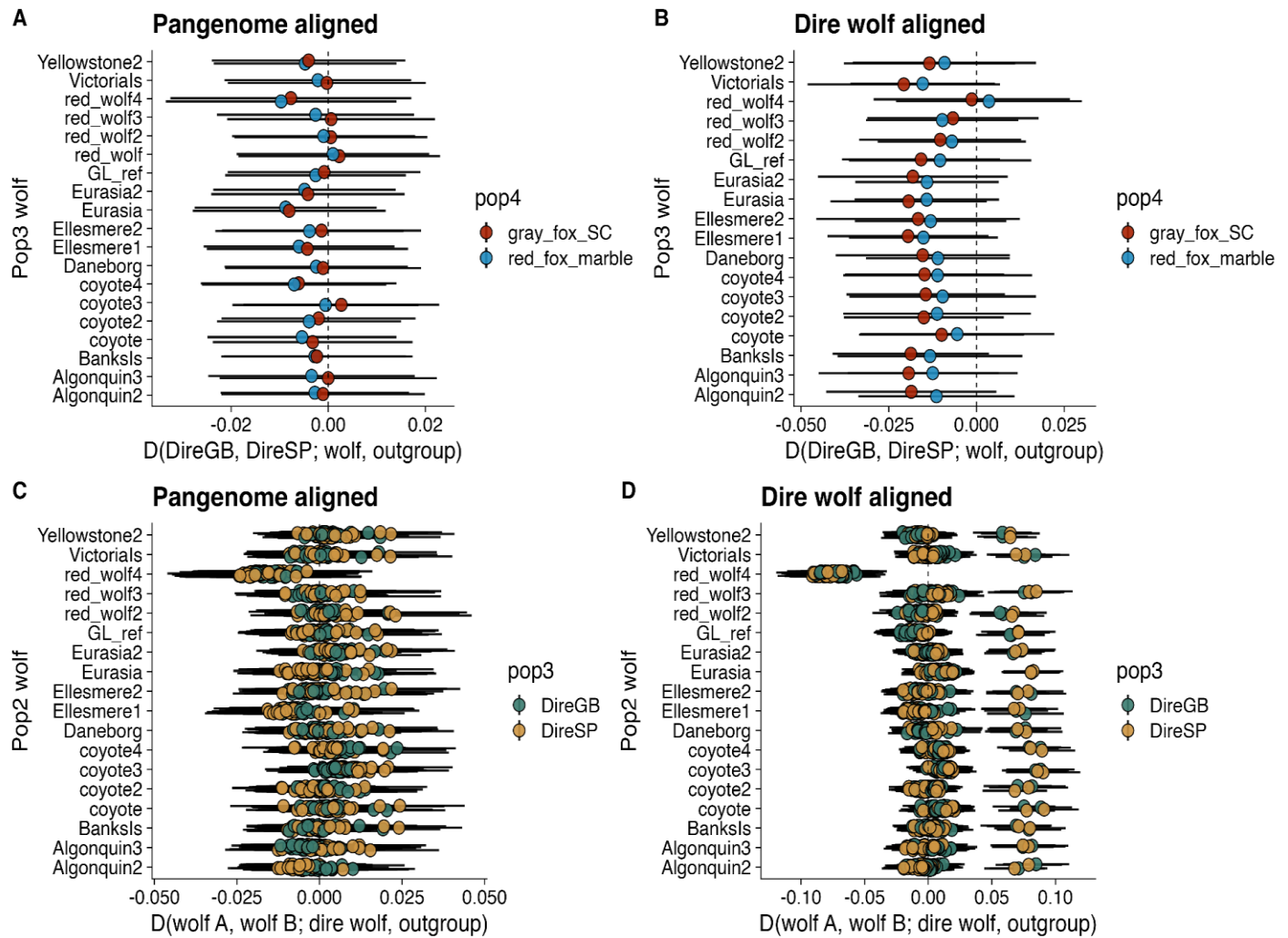
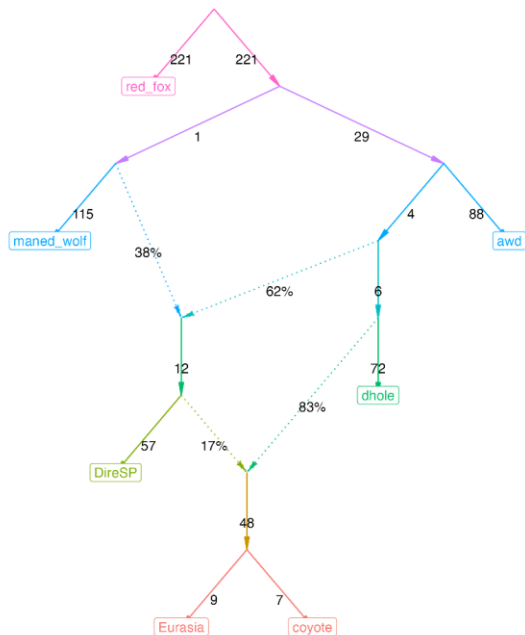


Figure S10. *D*-statistics testing whether the two dire wolf specimens, DireGB and DireSP, with various canid species, related to Figure 4. Each test examines whether the dire wolves form a clade relative to different individual wolves, coyotes and red wolves (**A,B**), and whether all wolves, coyotes, and red wolves form a clade relative to each dire wolf specimen (**C, D**). **A**) and **C**) show *D*-statistics ± 3 block-jackknife standard errors calculated using Consensify pseudohaploid genotypes from alignments to a canid pangenome graph, while **B**) and **D**) show *D*-statistics calculated using Consensify pseudohaploid genotypes from alignments to an iteratively polished dire wolf reference.

A score: 9.3; WR= 2.5



B score: 9.5; WR= 2.5

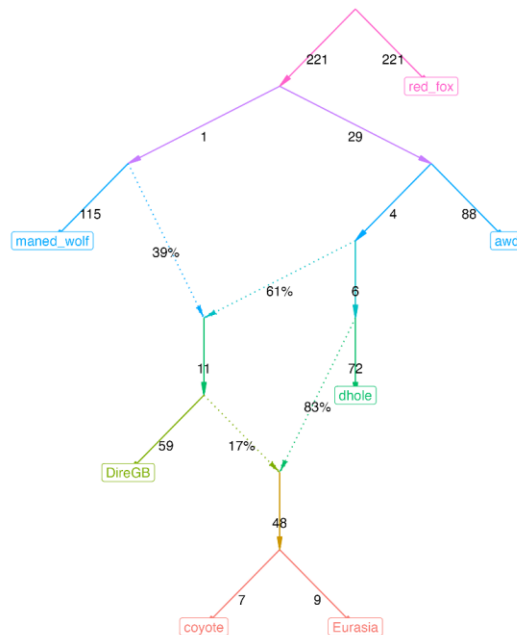
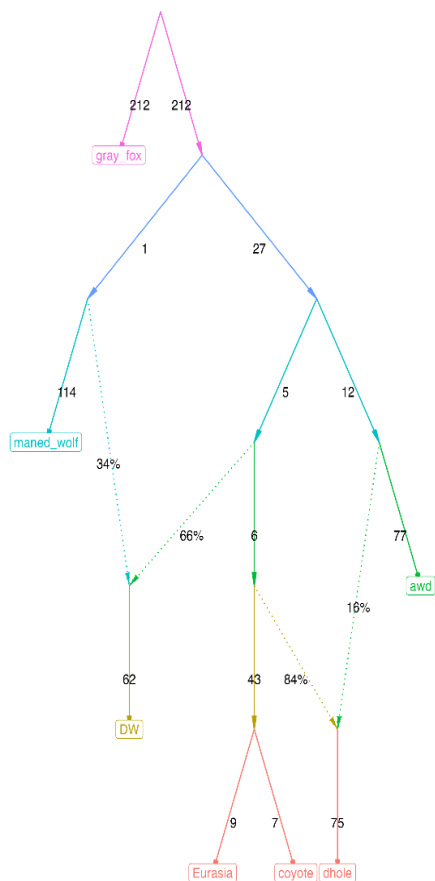
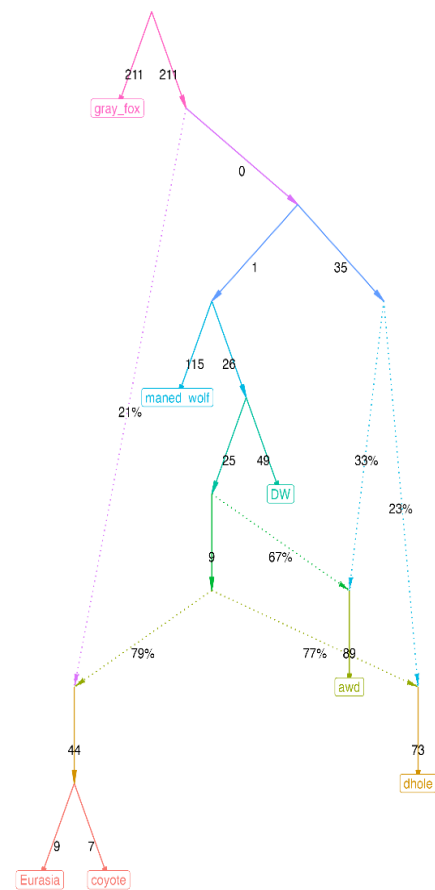


Figure S11. Best-fitting admixture graphs for DireSP (A) and DireGB (B), with the equivalent model grouping both dire wolf individuals, related to Figure 4. African wild dog is abbreviated (awd).

A score: 14.2; WR= 3.6



B score: 2.4; WR= 1.6



C score: 9.2; WR= 2.7

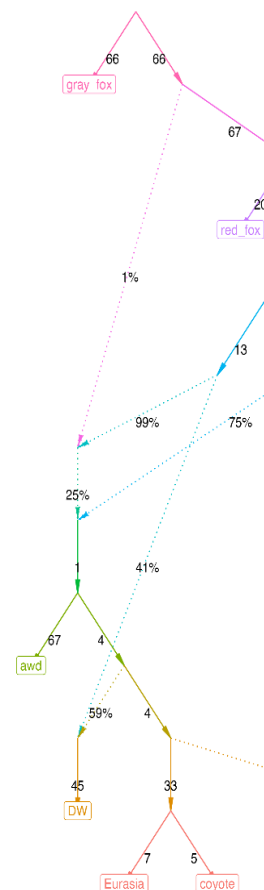
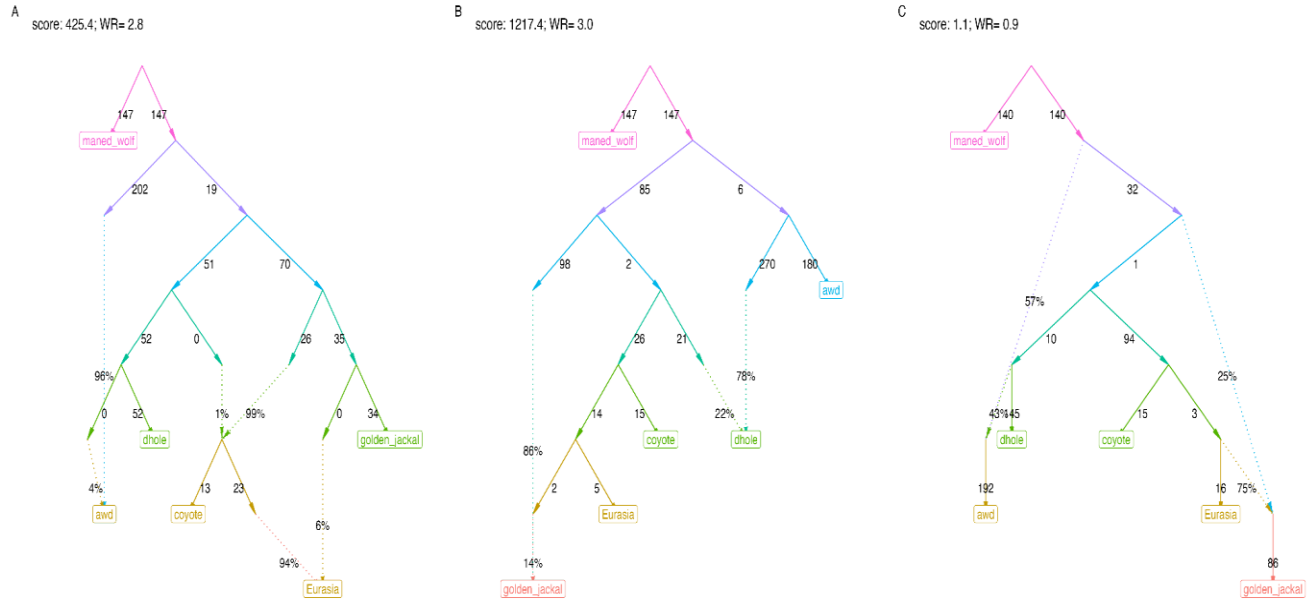


Figure S12 Best-fitting admixture graphs with gray fox as an outgroup, related to Figure 4. Fit with either **A)** two or **B)** three admixture events, and **C)** with the inclusion of red fox, a closer outgroup. Graphs were fit using Consensify pseudohaploid genotypes called from alignments to the canid pangenome. African wild dog is abbreviated (awd).



score: 11.5; WR= 2.7

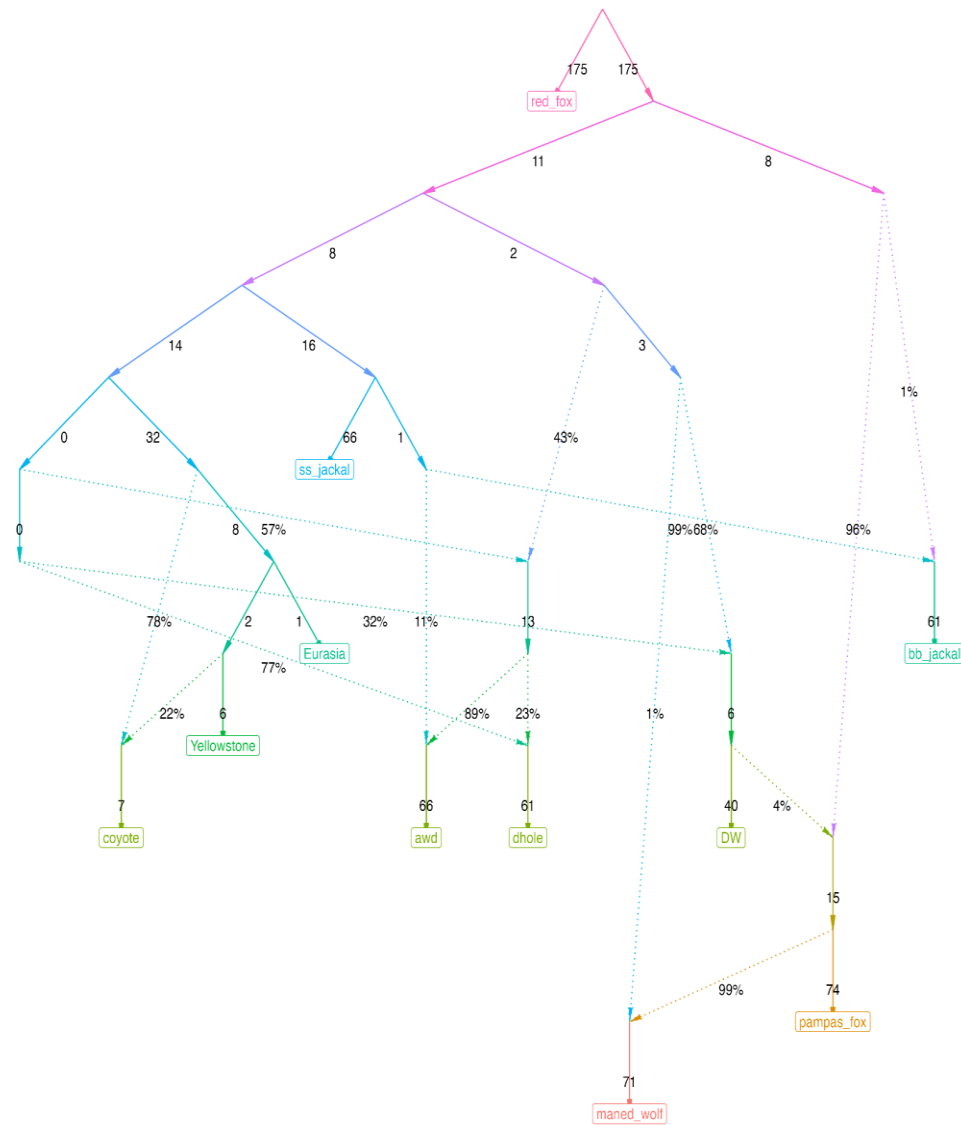


Figure S14. Best-fitting admixture graph model with 8 admixture events modeling population relationships across Canini, related to Figure 4. African wild dog is abbreviated (awd).

score: 24; WR= 3.6

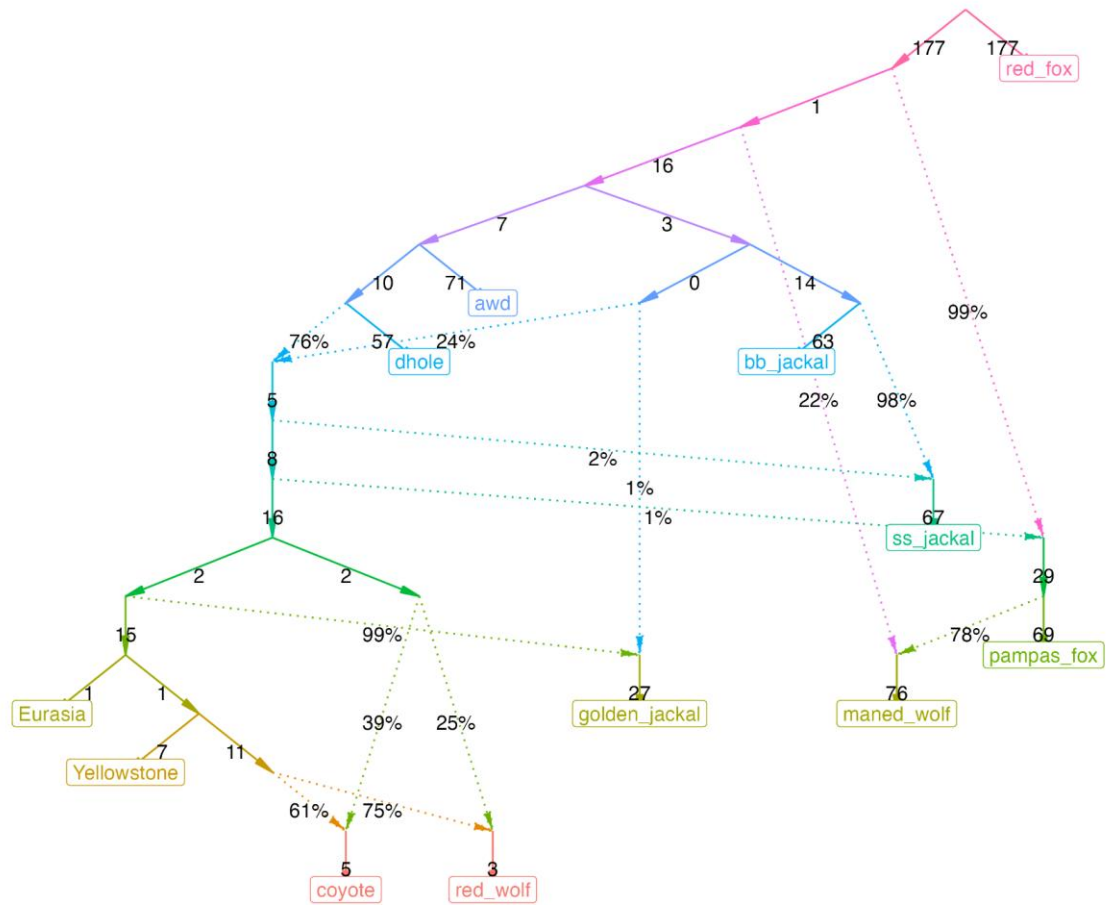


Figure S15. Best-fitting admixture graph with 7 admixture events which recovers gene flow events in species with known histories of admixture, related to Figure 4. Species include North American gray wolf ancestry in coyotes and red wolves and gene flow between gray wolves and golden jackals. African wild dog is abbreviated (awd).

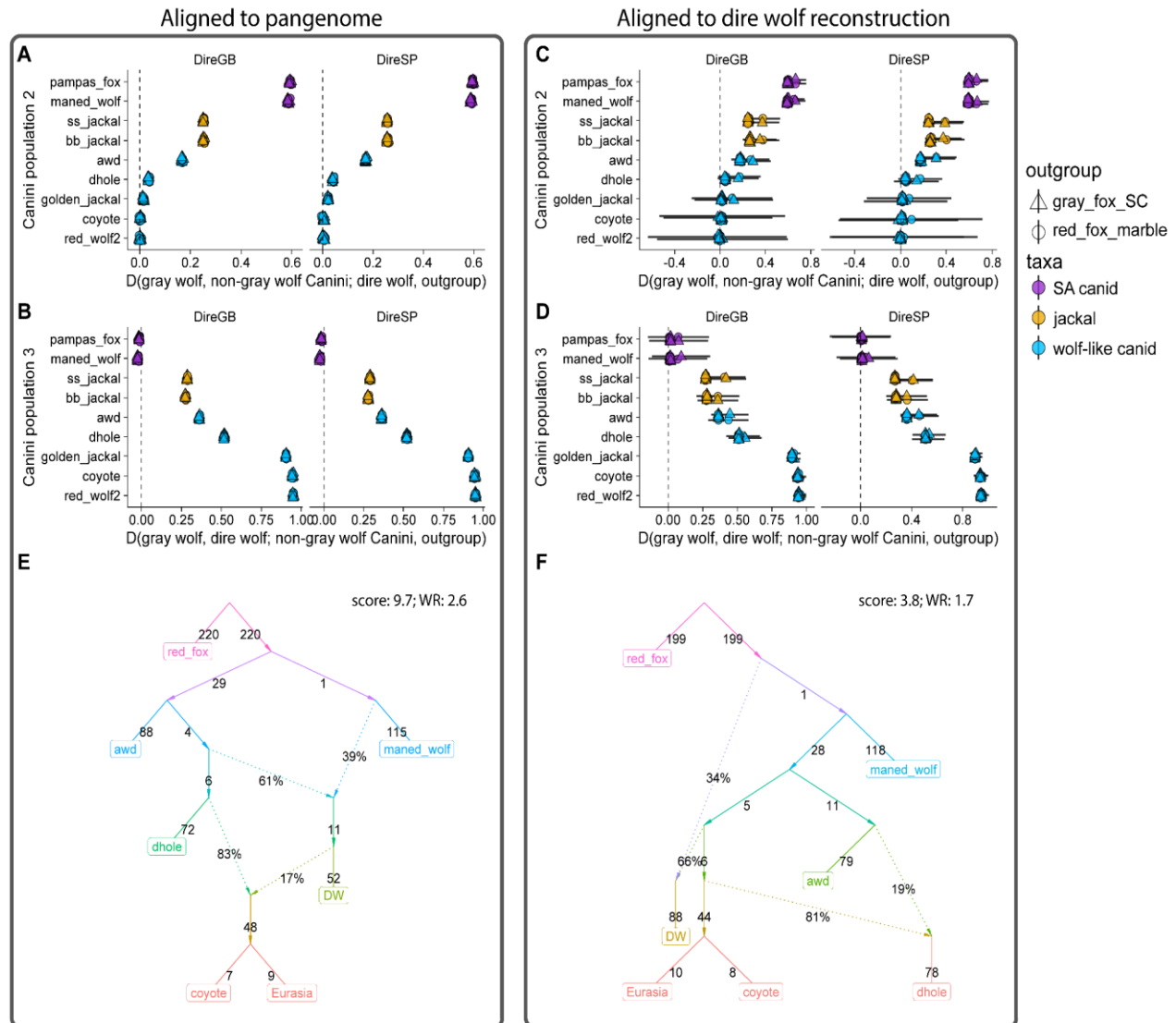


Figure S16. Comparison of admixture analyses using either a canid pangenome (A,B,E) or the dire wolf iterative genome (C, D, F) as a reference, related to Figure 4. A) and C) show allele sharing as D -statistics ± 3 block-jackknife standard errors between dire wolves and gray wolves, relative to other species within Canini. B) and D) show allele sharing between gray wolves and other Canini species, relative to dire wolves. E) and F) depict the best-fitting (lowest log-likelihood score) admixture graphs with two admixture events modeling the relationship between dire wolves and other canids. Both graphs model dire wolves with a mixture of basal and wolf-like canid ancestries, though the phylogenetic placement of the basal ancestry differs. The pangenome graph depicts the ancestors of coyotes and gray wolves as being admixed, while the dire wolf genome reference instead models dholes as admixed.

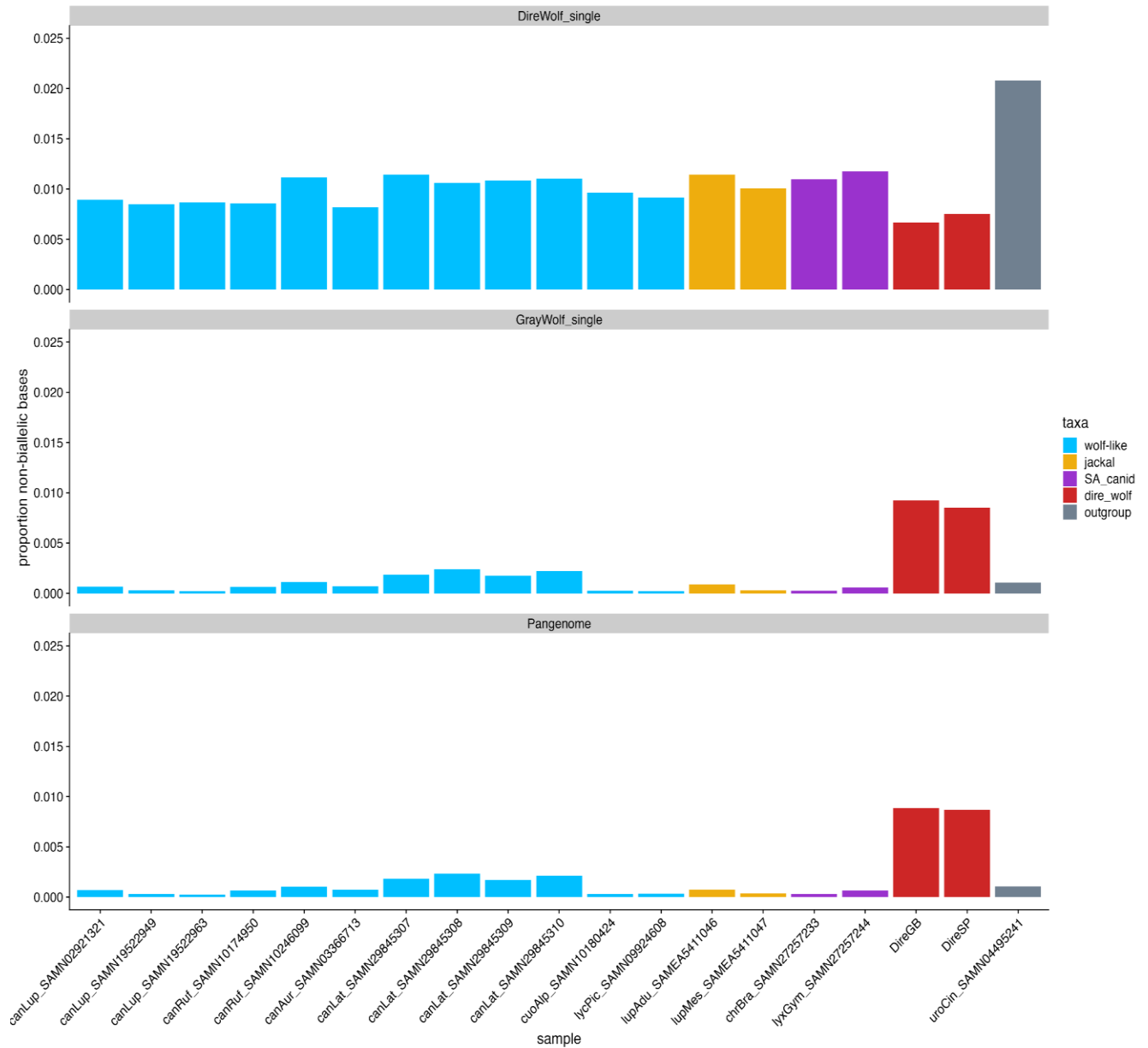


Figure S17. Rate of non-biallelic bases occurring in aligned reads at sites determined to be biallelic across canids, related to Figure 4. This is a proxy for relative error rate and reflects the rate of unexpected bases occurring due to either sequencing or mapping error, or ancient DNA damage. The rate of unexpected bases occurring is shown for reads mapped to the iteratively polished dire wolf genome (top), the gray wolf reference genome (mCanLor; middle), and a canid pangenome (bottom).

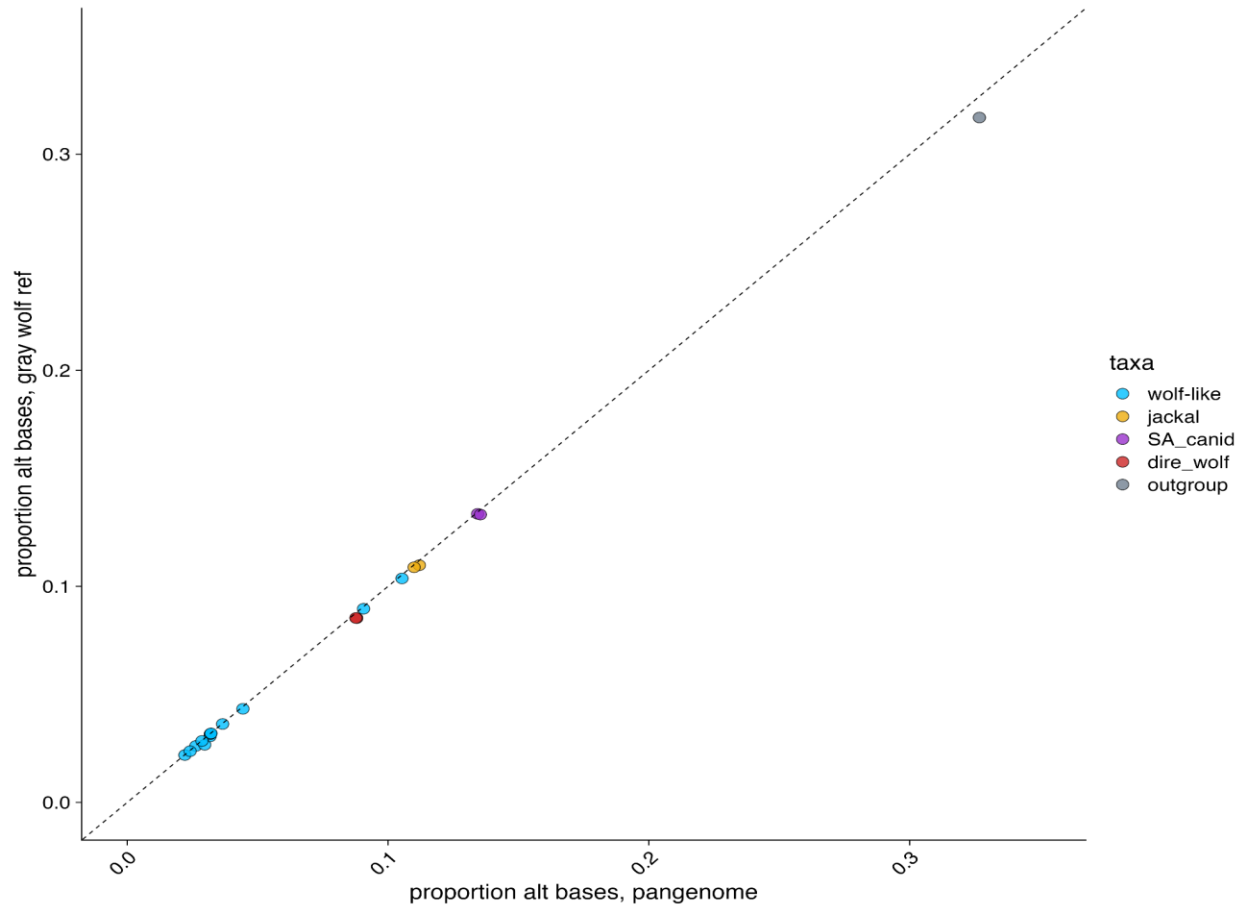


Figure S18. Proportion of bases carrying the alternative allele at biallelic sites for all canid individuals aligned either to a canid pangenome (x-axis) or to the gray wolf reference genome (y-axis), related to Figure 4. Pangenome alignments have a greater proportion of alternative alleles, especially for more divergent taxa.

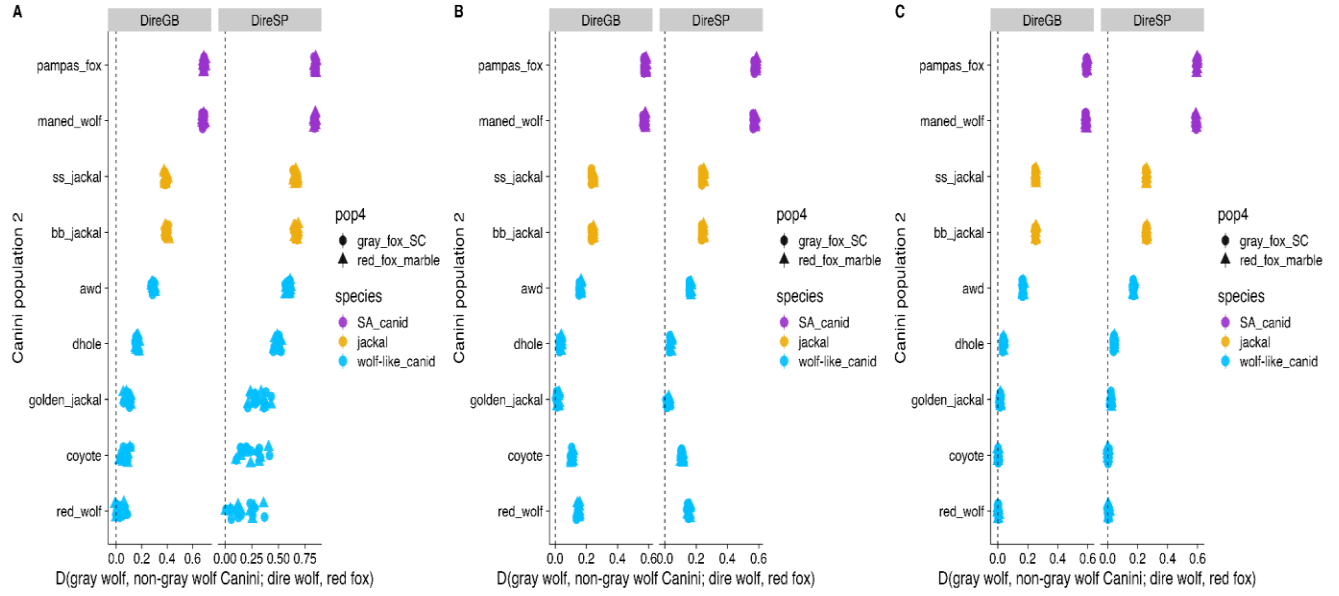


Figure S19: Comparison of D -statistics examining allele sharing between dire wolves and gray wolves calculated either using **A)** diploid genotypes called with *DeepVariant*, **B)** single-read pseudohaploid genotypes, or **C)** Consensify-style pseudohaploid genotypes, related to Figure 4. Diploid genotypes exhibit reference bias (excess allele sharing with gray wolves) which is exacerbated in the lower coverage SP individual, while single-read pseudohaploid genotypes have apparent outgroup attraction for taxa with higher sequencing error (coyote and red wolf, **Fig. S10**). The Consensify-style pseudohaploid genotypes appear to mitigate both sources of bias. African wild dog is abbreviated (awd).